

Construction of quotient schemes

SGA 3, Exposé V

English draft

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About this draft

This is an unofficial English translation of Exposé V, “Construction of quotient schemes” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp5-13oct24.pdf` (41 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé V. Construction of quotient schemes

by P. Gabriel

The object of this exposé is to prove the theorems stated in TDTE III.⁽¹⁾ If X and T are two objects of a category $\mathcal{C}$, we write $X(T)$ instead of $\text{Hom}_{\mathcal{C}}(T, X)$. Similarly, if $\varphi : Y \rightarrow X$ (resp. T) is an arrow (resp. an object) of $\mathcal{C}$, $\varphi(T)$ denotes the map $g \mapsto \varphi \circ g$ from $Y(T)$ to $X(T)$:

$$\begin{array}{ccc} & T & \\ g \swarrow & & \searrow \varphi \circ g \\ Y & \xrightarrow{\varphi} & X, \end{array}$$

and $T(\varphi)$ denotes the map $g \mapsto g \circ \varphi$ from $T(X)$ to $T(Y)$:

$$\begin{array}{ccc} Y & \xrightarrow{\varphi} & X \\ & \searrow g \circ \varphi & \swarrow g \\ & T. & \end{array}$$

Finally, if P is a scheme, one denotes by $\underline{P}$ the underlying set of P .

Exceptionally, in the present exposé we do not follow the convention stated in IV 4.6.15 concerning the notation for quotients (loc. cit., top of page 227 of the original), for we wish to give here a construction of quotients that also applies to “pre-equivalence relations”⁽²⁾ that are not equivalence relations.

1. $\mathcal{C}$ -groupoids

a) $\mathcal{C}$ is a category in which products and fibered products exist. Let us first recall that a diagram

$$X_1 \xrightleftharpoons[d_0]{d_1} X_0 \xrightarrow{p} Y$$

of $\mathcal{C}$ is said to be exact if $pd_0 = pd_1$ and if, for every $T \in \mathcal{C}$, $T(p)$ is a bijection from $T(Y)$ onto the subset of $T(X_0)$ consisting of the arrows $f : X_0 \rightarrow T$ such that $fd_0 = fd_1$. One also says that (Y, p) is the cokernel of (d_0, d_1) , and one writes

$$(Y, p) = \text{Coker}(d_0, d_1).$$

b) Let, for example, $\mathcal{C}$ be the category (Esp.An) of ringed spaces. In this case there is always a cokernel (Y, p) , of which one can give the following description: the underlying topological space of Y is obtained from X_0 by identifying the points $d_0(x)$ and $d_1(x)$ and giving Y the

⁽¹⁾**N.D.E.** Namely, Theorems 5.1, 5.3, 6.1, 6.2 and 7.2 of TDTE III. The first two (resp. the next two) correspond to Theorem 4.1 (resp. Theorems 7.1 and 8.1) of this exposé. Theorem 7.2 of TDTE III is proved in Exp. VIA, 3.2 and 3.3.

⁽²⁾**N.D.E.** That is, to groupoids with base X , cf. the terminology at the end of section 1. When $\mathcal{C}$ is the category of schemes, the quotient $p : X \rightarrow Y$ of a groupoid X_* with base X exists under certain hypotheses (cf. 4.1, 6.1, 7.1); if, moreover, X_* is an equivalence relation, p is, under the same hypotheses, faithfully flat and quasi-compact, hence a universal epimorphism, cf. loc. cit.

quotient topology. The canonical map $\pi : X_0 \rightarrow Y$ and d_0, d_1 then induce a pair of arrows of sheaves of rings on Y :

$$\pi_*(\mathcal{O}_0) \rightrightarrows^{\delta_0}_{\delta_1} \pi_*(d_{0*}\mathcal{O}_1) = \pi_*(d_{1*}\mathcal{O}_1),$$

where $\mathcal{O}_i$ is the structure sheaf of X_i . For the sheaf of rings on Y one chooses the subsheaf of $\pi_*(\mathcal{O}_0)$ whose sections s satisfy $\delta_0(s) = \delta_1(s)$. The arrow p is defined in the obvious way.

⁽³⁾Let $X_1 \rightrightarrows X_0$, with arrows d_1, d_0 , be a diagram in (Esp.An), and let (Y, p) be its cokernel. An open subset U of X_0 is said to be saturated if $d_0^{-1}(U) = d_1^{-1}(U)$, which is equivalent to saying that $U = p^{-1}(p(U))$. In this case, since Y has the quotient topology, $p(U)$ is an open subset of Y .

Lemma 1.1. *Let U be a saturated open subset of X and $V = p(U)$. If one denotes by U_1 the open subset $d_0^{-1}(U) = d_1^{-1}(U)$ of X_1 , and by $\tilde{d}_0, \tilde{d}_1$ and $\tilde{p}$ the restrictions of d_0, d_1 to U_1 , and of p to U , then $(V, \tilde{p})$ is a cokernel in (Esp.An) of:⁽⁴⁾*

$$U_1 \rightrightarrows^{\tilde{d}_1}_{\tilde{d}_0} U \xrightarrow{\tilde{p}} V.$$

The verification is easy.

Lemma 1.2. *Let $X_1 \rightrightarrows X_0$, with arrows d_1, d_0 , be a diagram in (Sch) and let (Y, p) be its cokernel in (Esp.An).*

- (i) *If Y is a scheme and p a morphism of schemes, then (Y, p) is a cokernel of (d_0, d_1) in (Sch).*
- (ii) *Suppose every point of X_0 has a saturated open neighborhood U such that, denoting by $\tilde{d}_0$ and $\tilde{d}_1$ the restrictions of d_0 and d_1 to $d_0^{-1}(U) = d_1^{-1}(U)$, and by (Q, q) the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An), Q is a scheme and q a morphism of schemes. Then (Y, p) is a cokernel of (d_0, d_1) in (Sch).*

(i) is proved in §4.c; since the proof is short, let us repeat it here. Let $f : X_0 \rightarrow Z$ be a morphism of schemes such that $fd_0 = fd_1$. By hypothesis there is one and only one morphism of ringed spaces $r : Y \rightarrow Z$ such that $f = rp$. One must show that, for every $y \in Y$, the homomorphism $\mathcal{O}_{r(y)} \rightarrow \mathcal{O}_y$ induced by r is local. This follows from the fact that p is surjective, hence y has the form $p(x)$, and that the homomorphism $\mathcal{O}_{f(x)} \rightarrow \mathcal{O}_x$ induced by f is local.

(ii) follows from (i) and the preceding lemma.

c) In this exposé we study the existence of $\text{Coker}(d_0, d_1)$ when the pair of arrows (d_0, d_1) is placed in a richer context; precisely, denote by $X_2 = X_1 \times_{d_1, d_0} X_1$ the fibered product of the diagram

$$(*) \quad \begin{array}{ccc} & X_1 & \\ & \downarrow d_1 & \\ X_1 & \xrightarrow{d_0} & X_0, \end{array}$$

⁽³⁾**N.D.E.** Lemmas 1.1 and 1.2, used several times in sections 5 to 9, have been added.

⁽⁴⁾**N.D.E.** This is not the case in the category of schemes. Let, for example, $S = \text{Spec}(\mathbf{C})$, $X_0 = \mathbb{A}_S^2 = \text{Spec}(\mathbf{C}[x_1, x_2])$, $d_1 : \mathbf{G}_{m,S} \times_S \mathbb{A}_S^2 \rightarrow \mathbb{A}_S^2$ the action of $\mathbf{G}_{m,S}$ by homotheties on $\mathbb{A}_S^2$, d_0 the projection onto the second factor, and $U = \mathbb{A}_S^2 - \{m\}$, where m is the point $(0, 0)$. Then the projective space $\mathbb{P}_S^1$ is the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An) and in (Sch), and the cokernel Y of (d_0, d_1) in (Esp.An) is the union of $\mathbb{P}_S^1$ and the point $y_0 = \{p(m)\}$; the only open subset containing y_0 is Y , and one has $\Gamma(Y, \mathcal{O}_Y) = \mathbf{C}$. If $f : \mathbb{A}_S^2 \rightarrow T$ is a morphism of S -schemes such that $fd_0 = fd_1$, and if $V = \text{Spec}(A)$ is an affine open subset of T containing the point $t_0 = f(y_0)$, then $f^{-1}(V) = \mathbb{A}^2$ and the ring homomorphism $A \rightarrow \mathbf{C}[x_1, x_2]$ factors through $\mathbf{C}$; this shows that $S = \text{Spec}(\mathbf{C})$ is the cokernel of (d_0, d_1) in the category (Sch/ S).

and by d'_0 and d'_2 the two canonical projections of X_2 onto X_1 ; one therefore has, by definition, a cartesian square

$$(0) \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_0} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_0} & X_0. \end{array}$$

Moreover, give oneself a third arrow $d'_1 : X_2 \rightarrow X_1$; we say that $(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$ is a $\mathcal{C}$ -groupoid if for every object T of $\mathcal{C}$, $X_1(T)$ is the set of arrows of a groupoid $X_*(T)$ whose set of objects is $X_0(T)$, whose source map is $d_1(T)$, whose target map is $d_0(T)$, and whose composition map is $d'_1(T)$ (as usual, one identifies $(X_1 \times_{d_1, d_0} X_1)(T)$ with $X_1(T) \times_{d_1(T), d_0(T)} X_1(T)$; one also recalls that a groupoid is a category all of whose arrows are invertible).⁽⁵⁾

If φ is an arrow of the groupoid $X_*(T)$, the map $f \mapsto \varphi \circ f$ is a bijection from the set of arrows f whose target coincides with the source of φ onto the set of arrows having the same target as φ . One easily sees that this fact can be expressed by saying that the square

$$(1) \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ d'_0 \downarrow & & \downarrow d_0 \\ X_1 & \xrightarrow{d_0} & X_0 \end{array}$$

is cartesian.

Similarly, the map $g \mapsto g \circ \varphi$ is a bijection from the set of arrows g of $X_*(T)$ whose source is the target of φ onto the set of arrows having the same source as φ . One can again express this fact by saying that the square

$$(2) \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_1} & X_0 \end{array}$$

is cartesian.

On the other hand, let $s : X_0 \rightarrow X_1$ be the unique arrow of $\mathcal{C}$ such that, for every T , $s(T) : X_0(T) \rightarrow X_1(T)$ assigns to every object of $X_*(T)$ its identity arrow.⁽⁶⁾ The arrow s satisfies the equalities

$$(3) \quad d_1 s = \text{id}_{X_0},$$

and

$$(3 \text{ bis}) \quad d_0 s = \text{id}_{X_0}.$$

Finally, associativity of the composition maps $d'_1(T)$ is expressed by commutativity of the diagram

$$(4) \quad \begin{array}{ccc} X_1 \times_{d_1, d_0} X_1 \times_{d_1, d_0} X_1 & \xrightarrow{d'_1 \times X_1} & X_1 \times_{d_1, d_0} X_1 \\ X_1 \times d'_1 \downarrow & & \downarrow d'_1 \\ X_1 \times_{d_1, d_0} X_1 & \xrightarrow{d'_1} & X_1. \end{array}$$

⁽⁵⁾**N.D.E.** Thus, in this case, $X_2(T)$ is the set of pairs (f_2, f_1) of composable arrows, that is, such that $d_0(f_1) = d_1(f_2)$, and d'_0, d'_1, d'_2 send (f_2, f_1) to $f_2, f_2 \circ f_1, f_1$ respectively.

⁽⁶⁾**N.D.E.** $T \mapsto s(T)$ defines an element of $\text{Hom}(h_{X_0}, h_{X_1})$, and the latter equals $\text{Hom}(X_0, X_1)$ by Yoneda's lemma.

Conversely, conditions (1), (2) and (4) and the existence of an arrow s satisfying (3) imply that $(X_1 \rightrightarrows X_0, d'_1)$, with arrows d_1, d_0 , is a $\mathcal{C}$ -groupoid. Condition (3) is harmless; it simply ensures that the map $d_1(T) : X_1(T) \rightarrow X_0(T)$ is surjective for every $T \in \mathcal{C}$. In the remainder of this exposé we chiefly use the cartesian squares (0), (1) and (2), which we summarize in the diagram

$$(0,1,2) \quad \begin{array}{ccccc} X_2 & \xrightleftharpoons[d'_0]{d'_1} & X_1 & \xrightarrow{d_0} & X_0 \\ d'_2 \downarrow & & \downarrow d_1 & & \\ X_1 & \xrightleftharpoons[d_0]{d_1} & X_0 & & \end{array} .$$

In this diagram the two squares on the left (that is, squares (0) and (2)) are cartesian; the first row is exact and X_2 is identified with the fibered product $X_1 \times_{d_0, d_0} X_1$.

We use associativity only indirectly, for example to ensure the existence of an arrow s satisfying (3) and (3 bis), or to ensure the existence of an arrow

$$(\dagger) \quad \sigma : X_1 \rightarrow X_1 \quad \text{such that} \quad d_0 \sigma = d_1 \text{ and } d_1 \sigma = d_0$$

(one chooses σ so that $\sigma(T) : X_1(T) \rightarrow X_1(T)$ sends each arrow of $X_*(T)$ to its inverse arrow).⁽⁷⁾

By abuse of language, we shall sometimes call a diagram

$$X_2 \xrightleftharpoons[d_0]{d'_0, d'_1, d'_2} X_1 \xrightleftharpoons[d_0]{d_0, d_1} X_0$$

a $\mathcal{C}$ -groupoid if (0), (1) and (2) are cartesian, (4) is commutative and there exists s satisfying (3). The object X_2 may thus be “a” fibered product of $(*)$ without being “the” fibered product of $(*)$.⁽⁸⁾

Terminology. Instead of the $\mathcal{C}$ -groupoid X_* , we shall also speak of the groupoid X_* with base X_0 , or of the pre-equivalence relation X_* in X_0 .

2. Examples of $\mathcal{C}$ -groupoids

a) Let X be an object of $\mathcal{C}$ and G a $\mathcal{C}$ -group acting on X on the left. We denote by $d_0 : G \times X \rightarrow X$ the arrow defining the action of G on X , by $d_1 : G \times X \rightarrow X$ the projection of the product onto the second factor, by $\mu : G \times G \rightarrow G$ the arrow defining the $\mathcal{C}$ -group structure of G , and finally by $\text{pr}_{2,3}$ the projection of $G \times G \times X = G \times (G \times X)$ onto the second factor. Then

$$G \times G \times X \xrightleftharpoons[G \times d_0]{\text{pr}_{2,3} \atop \rightarrow \mu \times X} G \times X \xrightleftharpoons[d_0]{d_1} X$$

is a $\mathcal{C}$ -groupoid.

b) Let $d_0, d_1 : X_1 \rightarrow X_0$ be an equivalence pair, that is, if $d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times X_0$ is the arrow with components d_0 and d_1 , we suppose that $(d_0 \boxtimes d_1)(T)$ is, for every object T of $\mathcal{C}$, a bijection from $X_1(T)$ onto the graph of an equivalence relation on $X_0(T)$. The set $X_1(T)$ is consequently identified with the set of pairs (x, y) of elements of $X_0(T)$ such that $x \sim y$; similarly, the set $X_2(T) = (X_1 \times_{d_1, d_0} X_1)(T)$ is identified with the set of triples (x, y, z) of elements of $X_0(T)$ such that $x \sim y$ and $y \sim z$. There is therefore one and only one arrow $d'_1 : X_2 \rightarrow X_1$ making squares (1) and (2) commutative: $d'_1(T)$ must send $(x, y, z) \in X_2(T)$ to $(x, z) \in X_1(T)$. For this choice of d'_1 , $(d_0, d_1 : X_1 \rightrightarrows X_0, d'_1)$ is a $\mathcal{C}$ -groupoid.

⁽⁷⁾**N.D.E.** It follows from Yoneda’s lemma that σ is an involutive automorphism of X_1 ; this will be used, for example, in 3.e) and in Theorem 4.1.

⁽⁸⁾**N.D.E.** See example 2.a) below.

Conversely, consider a $\mathcal{C}$ -groupoid X_* such that $d_0 \boxtimes d_1 : X_1 \longrightarrow X_0 \times X_0$ is a monomorphism. Then (d_0, d_1) is an equivalence pair, and X_* can be reconstructed from (d_0, d_1) as explained a few lines above.⁽⁹⁾

c) If $p : X \longrightarrow Y$ is any arrow of $\mathcal{C}$, and if pr_1 and pr_2 are the two projections of $X \times_{p,p} X$ onto X , then $(\text{pr}_1, \text{pr}_2) : X \times_{p,p} X \rightrightarrows X$ is an equivalence pair. One says that p is an effective epimorphism if the diagram

$$X \times_{p,p} X \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xrightarrow{\text{pr}_2} \end{array} X \xrightarrow{p} Y$$

is exact, that is, if $(Y, p) = \text{Coker}(\text{pr}_1, \text{pr}_2)$.

Let, for example, S be a noetherian scheme and $\mathcal{C}$ the category of schemes finite over S . Let us show that an epimorphism of $\mathcal{C}$ is not necessarily effective: one chooses $S = \text{Spec } k[T^3, T^5]$, where k is a commutative field, $Y = S$ and $X = \text{Spec } k[T]$. If i is the inclusion of $B = k[T^3, T^5]$ into $A = k[T]$, p is chosen to be $\text{Spec } i$. In this case $X \times_{p,p} X$ is identified with $\text{Spec}(A \otimes_B A)$ and $\text{Coker}(\text{pr}_1, \text{pr}_2)$ with $\text{Spec } B'$, where B' is the subring of A consisting of the a such that $a \otimes_B 1 = 1 \otimes_B a$. Now

$$T^7 \otimes_B 1 = (T^2 T^5) \otimes_B 1 = T^2 \otimes_B T^5 = T^2 \otimes_B (T^3 T^2) = T^5 \otimes_B T^2 = 1 \otimes_B T^7.$$

Thus T^7 belongs to B' , does not belong to B , and $\text{Spec } B'$ is distinct from $\text{Spec } B$, whence the counterexample.⁽¹⁰⁾

3. Some sorites on $\mathcal{C}$ -groupoids

Here, in no particular order, are some remarks used below:

a) Let

$$X_2 \begin{array}{c} \xrightarrow{d'_0, d'_1, d'_2} \\ \xrightarrow{d_0, d_1} \end{array} X_1 \xrightarrow{d_0, d_1} X_0$$

be a $\mathcal{C}$ -groupoid and $f_0 : Y_0 \longrightarrow X_0$ an arrow of $\mathcal{C}$. We shall define a $\mathcal{C}$ -groupoid with base Y_0

$$Y_2 \begin{array}{c} \xrightarrow{e'_0, e'_1, e'_2} \\ \xrightarrow{e_0, e_1} \end{array} Y_1 \xrightarrow{e_0, e_1} Y_0$$

which will be said to be induced by X_* and f_0 . One will also say that Y_* is the inverse image of X_* by the base change f_0 .

For Y_1 we choose the fibered product of the diagram

$$\begin{array}{ccc} Y_1 & \xrightarrow{f_1} & X_1 \\ \downarrow & & \downarrow d_0 \boxtimes d_1 \\ Y_0 \times Y_0 & \xrightarrow{f_0 \times f_0} & X_0 \times X_0, \end{array}$$

and for e_0 and e_1 the arrows obtained by composing the canonical arrow $Y_1 \longrightarrow Y_0 \times Y_0$ with the first and second projections of $Y_0 \times Y_0$. The morphism $Y_1 \longrightarrow Y_0 \times Y_0$ is then $e_0 \boxtimes e_1$, and one has $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, where f_1 denotes the projection of Y_1 onto X_1 .

One puts $Y_2 = Y_1 \times_{e_0, e_1} Y_1$, cf. 1.c). One can say that the pair (e_0, e_1) is defined so that, for every $T \in \mathcal{C}$, and for every pair (y, x) of elements of $Y_0(T)$, there is a certain one-to-one correspondence $\psi \longmapsto {}_y\psi_x$ between the arrows ψ of $X_*(T)$ with source $f_0(x)$ and target $f_0(y)$, and the arrows ${}_y\psi_x$ of $Y_*(T)$ with source x and target y . One therefore determines $e'_1 : Y_2 \longrightarrow Y_1$ by defining, for every $T \in \mathcal{C}$, the composition of arrows of $Y_*(T)$ by the formula

$$z\psi_y \circ {}_y\varphi_x = z(\psi \circ \varphi)_x.$$

⁽⁹⁾**N.D.E.** In particular, if G is a $\mathcal{C}$ -group acting on the left on an object X of $\mathcal{C}$ and if X_* is the $\mathcal{C}$ -groupoid defined in a), then (d_0, d_1) is an equivalence pair if and only if G acts freely on X , cf. Exp. III, 3.2.1.

⁽¹⁰⁾**N.D.E.** The same argument applies to $B = k[T^3, T^4]$ and $T^5 \otimes_B 1$; more generally, to $B = k[T^n, T^{n+r}]$ and the element $T^{n+2r} \otimes_B 1$, provided n does not divide $2r$.

It is clear that this definition makes each $Y_*(T)$ a groupoid.

b) Knowing the $\mathcal{C}$ -groupoid X_* and the base change $f_0 : Y_0 \rightarrow X_0$, one can reconstruct the pair $(e_0, e_1) : Y_1 \rightrightarrows Y_0$ in another way:⁽¹¹⁾ construct $Y_0 \times_{X_0} X_1$, pr_1 and pr_2 so that the square

$$\begin{array}{ccc} Y_0 \times_{X_0} X_1 & \xrightarrow{\text{pr}_2} & X_1 \\ \text{pr}_1 \downarrow & & \downarrow d_0 \\ Y_0 & \xrightarrow{f_0} & X_0 \end{array}$$

is cartesian. One then verifies without difficulty, by reduction to the set-theoretic case, that one has the cartesian square:

$$\begin{array}{ccc} Y_1 & \xrightarrow{e_0 \boxtimes f_1} & Y_0 \times_{X_0} X_1 \\ e_1 \downarrow & & \downarrow d_1 \circ \text{pr}_2 \\ Y_0 & \xrightarrow{f_0} & X_0, \end{array}$$

where f_1 denotes the canonical projection of $Y_1 = (Y_0 \times Y_0) \times_{(X_0 \times X_0)} X_1$ onto X_1 .

c) We shall give two examples of inverse images of a $\mathcal{C}$ -groupoid. Take $Y_0 = X_1$, $f_0 = d_0$. For every object T of $\mathcal{C}$, $Y_1(T)$ is then identified with the set of diagrams of the form

$$\begin{array}{ccc} b & \xrightarrow{\varphi} & d \\ f \uparrow & & \uparrow g \\ a & & c \end{array}$$

of $X_*(T)$. The source of such a diagram is the arrow f , the target is the arrow g . These diagrams compose in the obvious way.

Now put $Y'_0 = X_1$, $f'_0 = d_1$ (we add primes⁽¹²⁾ to avoid any confusion with the preceding example). In this case, for every $T \in \mathcal{C}$, $Y'_1(T)$ is identified with the set of diagrams of the form

$$\begin{array}{ccc} b & & d \\ f \uparrow & & \uparrow g \\ a & \xrightarrow{\psi} & c \end{array}$$

of the groupoid $X_*(T)$. The source of such a diagram is f , the target is g ; the composition of these diagrams is obvious.

This being said, it is clear that the identity map of $Y_0(T)$ and the map

$$\begin{array}{ccc} b & \xrightarrow{\varphi} & d \\ f \uparrow & & \uparrow g \\ a & & c \end{array} \quad \mapsto \quad \begin{array}{ccc} b & & d \\ f \uparrow & & \uparrow g \\ a & \xrightarrow{g^{-1}\varphi f} & c \end{array}$$

from $Y_1(T)$ onto $Y'_1(T)$ define an isomorphism of the groupoid $Y_*(T)$ onto $Y'_*(T)$. Moreover, this isomorphism depends functorially on T , so that the $\mathcal{C}$ -groupoids Y_* and Y'_* are isomorphic.⁽¹³⁾

d)

⁽¹¹⁾**N.D.E.** This second viewpoint will be used in 3.f) and in the proof of 6.1.

⁽¹²⁾**N.D.E.** “Accents” in the original.

⁽¹³⁾**N.D.E.** This will play a crucial role in the proof of Lemma 6.1.

Proposition 3.1. *We retain the notation of a), and suppose that f_0 is an effective and universal epimorphism. Then $\text{Coker}(d_0, d_1)$ exists if and only if $\text{Coker}(e_0, e_1)$ exists.⁽¹⁴⁾ Moreover, in this case f_0 induces an isomorphism*

$$\text{Coker}(d_0, d_1) \xrightarrow{\sim} \text{Coker}(e_0, e_1).$$

Let us first recall that an epimorphism $f_0 : Y_0 \rightarrow X_0$ is said to be universal if, for every cartesian square

$$\begin{array}{ccc} Y' & \longrightarrow & Y_0 \\ f' \downarrow & & \downarrow f_0 \\ X' & \longrightarrow & X_0, \end{array}$$

f' is an epimorphism. This being said, denote by $C(d_0, d_1)$ the covariant functor from $\mathcal{C}$ to sets which assigns to every $T \in \mathcal{C}$ the kernel of the pair $T(d_0), T(d_1) : T(X_0) \rightrightarrows T(X_1)$. Define $C(e_0, e_1)$ similarly. For every $T \in \mathcal{C}$, one therefore has a commutative diagram

$$\begin{array}{ccccc} C(d_0, d_1)(T) & \longrightarrow & T(X_0) & \xrightleftharpoons[T(d_0)]{T(d_1)} & T(X_1) \\ T(f) \downarrow & & T(f_0) \downarrow & & \downarrow T(f_1) \\ C(e_0, e_1)(T) & \longrightarrow & T(Y_0) & \xrightleftharpoons[T(e_0)]{T(e_1)} & T(Y_1), \end{array}$$

where $T(f)$ is the injection induced by the injection $T(f_0)$. If we show that $T(f)$ is a surjection for every T , we shall have a functorial isomorphism $f : C(d_0, d_1) \xrightarrow{\sim} C(e_0, e_1)$, so that the representability of one of these functors is equivalent to that of the other; this will prove our proposition.

To prove surjectivity of $T(f)$, consider the diagram

$$\begin{array}{ccccc} & & Y_1 & \xrightarrow{f_1} & X_1 \\ & \Delta \nearrow & \downarrow e_1 & \downarrow e_0 & \downarrow d_1 \\ Y_0 \times_{X_0} Y_0 & \xrightarrow[\text{pr}_1]{\text{pr}_2} & Y_0 & \xrightarrow{f_0} & X_0, \end{array}$$

where Δ is the section of $Y_1 \rightarrow Y_0 \times Y_0$ defined by the morphism $s \circ f_0 \circ \text{pr}_1 : Y_0 \times Y_0 \rightarrow X_1$, the arrow $s : X_0 \rightarrow X_1$ satisfying equalities (3) and (3 bis) of paragraph 1.

If the arrow $g : Y_0 \rightarrow T$ satisfies $g \circ e_0 = g \circ e_1$, one has $g \circ e_0 \circ \Delta = g \circ e_1 \circ \Delta$, hence $g \circ \text{pr}_1 = g \circ \text{pr}_2$. Since f_0 is an effective epimorphism, g is the composite of f_0 with an arrow $h : X_0 \rightarrow T$, that is, one has $g = T(f_0)(h)$. It remains to show that h belongs to $C(d_0, d_1)(T)$, that is, satisfies $hd_0 = hd_1$; now one has

$$hd_0f_1 = hf_0e_0 = ge_0 = ge_1 = hf_0e_1 = hd_1f_1,$$

whence the desired equality, since f_1 is an epimorphism (for f_0 is a universal epimorphism).

e) Now consider a scheme S and choose $\mathcal{C} = (\text{Sch}/S)$. Giving a $\mathcal{C}$ -groupoid

$$X_2 \xrightleftharpoons[d'_0]{d'_1} X_1 \xrightleftharpoons[d_0]{d_1} X_0$$

allows one to define an equivalence relation on the underlying set $\underline{X}_0$ of the scheme X_0 : if $x, y \in \underline{X}_0$, one writes $x \sim y$ when there exists $z \in \underline{X}_1$ such that $x = d_1z$ and $y = d_0z$. Reflexivity and symmetry of this relation are obvious;⁽¹⁵⁾ let us prove transitivity: if $x \sim y$

⁽¹⁴⁾**N.D.E.** The original has been modified to bring out the isomorphism below.

⁽¹⁵⁾**N.D.E.** Reflexivity follows from the existence of $s : X_0 \rightarrow X_1$, which is a section of d_0 and d_1 ; symmetry follows from the existence of the involution σ of X_1 which “interchanges d_0 and d_1 ”, that is, satisfies $d_0\sigma = d_1$ and $d_1\sigma = d_0$, cf. §1, (3), (3 bis) and (†).

and $y \sim z$, there exist $u, v \in \underline{X}_1$ such that $x = d_1 u$, $y = d_0 u$, $y = d_1 v$, $z = d_0 v$. It follows that (v, u) belongs to the set-theoretic fibered product $\underline{X}_1 \times_{d_1, d_0} \underline{X}_1$. Since the canonical map

$$\underline{X}_1 \times_{d_1, d_0} \underline{X}_1 \longrightarrow \underline{X}_1 \times_{d_1, d_0} \underline{X}_1$$

from the underlying set of the fibered product to the fibered product of the underlying sets is surjective, (v, u) is the image of some $w \in \underline{X}_2$. One then has $x = d_1 d'_1 w$ and $z = d_0 d'_1 w$, whence $x \sim z$.

f) Retain the notation of a) and b), $\mathcal{C}$ still being (Sch/S) . If x, y are points of Y_0 , we shall see that $x \sim y$ if and only if $f_0(x) \sim f_0(y)$ (the inverse image of the equivalence relation defined by a groupoid is the equivalence relation defined by the inverse image of the groupoid).

Indeed, suppose $x \sim y$. There thus exists $z \in \underline{Y}_1$ such that $x = e_1(z)$ and $y = e_0(z)$. Since $f_0 \circ e_i = d_i \circ f_1$ for $i = 0, 1$, one then has $f_0(x) = d_1 f_1(z)$ and $f_0(y) = d_0 f_1(z)$, whence $f_0(x) \sim f_0(y)$.

Conversely, suppose $f_0(x) \sim f_0(y)$, and let $z \in \underline{X}_1$ satisfy $f_0(y) = d_1(z)$ and $f_0(x) = d_0(z)$. Using the construction and notation of b), there is then a point t of $Y_0 \times_{X_0} X_1$ such that $\text{pr}_1(t) = x$ and $\text{pr}_2(t) = z$. Similarly, since $f_0(y) = d_1 \text{pr}_2(t)$, there is an $s \in \underline{Y}_1$ such that $y = e_1(s)$ and $(e_0 \boxtimes f_1)(s) = t$. One then has $e_0(s) = \text{pr}_1(e_0 \boxtimes f_1)(s) = \text{pr}_1(t) = x$. Whence $x \sim y$.

4. Passage to the quotient by a finite flat groupoid (proof of a special case)

Theorem 4.1. *Consider a (Sch/S) -groupoid:*

$$X_2 \begin{array}{c} \xrightarrow{d'_1} \\ \xrightarrow{d'_0} \end{array} X_1 \xrightarrow{d_0} X_0$$

Suppose the following conditions hold.⁽¹⁶⁾

- a) d_1 is finite locally free,
- b) for every $x \in X_0$, the set $d_0 d_1^{-1}(x)$ is contained in an affine open subset of X_0 .⁽¹⁷⁾

Then:

- (i) There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S) ; moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.
- (ii) p is integral and open, and Y is affine if X_0 is affine.⁽¹⁸⁾
- (iii) The morphism $X_1 \longrightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.
- (iv) If (d_0, d_1) is an equivalence pair, $X_1 \longrightarrow X_0 \times_Y X_0$ is an isomorphism⁽¹⁹⁾ and $p : X_0 \longrightarrow Y$ is finite locally free.⁽²⁰⁾ Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology and, for every base change $Y' \longrightarrow Y$, Y' is the cokernel of the groupoid $X_* \times_Y Y'$ obtained from X_* by the base change $X_0 \times_Y Y' \longrightarrow X_0$.

In particular, for every base change $S' \longrightarrow S$, $Y' = Y \times_S S'$ is the cokernel of the S' -groupoid $X'_* = X_* \times_S S'$. Thus, in this case, “formation of the quotient commutes with base change”.

⁽¹⁶⁾**N.D.E.** Since $d_0 = d_1 \sigma$, where σ is an involutive automorphism of X_1 , these two conditions are symmetric in d_1 and d_0 ; moreover one has $d_0 d_1^{-1}(x) = d_1 d_0^{-1}(x)$.

⁽¹⁷⁾**N.D.E.** Hypothesis b) cannot be omitted. Indeed, H. Hironaka gave an example of an action of the finite group $G = \mathbf{Z}/2\mathbf{Z}$ on a proper $\mathbf{C}$ -variety X_0 , such that the quotient X_0/G is an algebraic space which is not a scheme ([Hi62], see also [Mum65], Chap. 4, §3).

⁽¹⁸⁾**N.D.E.** The assertion that p is open has been added, using the analogous proof given in 6.1.

⁽¹⁹⁾**N.D.E.** Note that, in this case, $X_1 \longrightarrow X_0 \times_S X_0$ is therefore an immersion (EGA I, 5.3.10); see also VIB, 9.2.1. On the other hand, for the existence of the quotient (in the category of schemes or that of algebraic spaces) under the weaker hypothesis that d_0 and d_1 are finite (but not necessarily flat), see [An73], §1.1, [Fe03], [Ko08]. . .

⁽²⁰⁾**N.D.E.** The following consequences have been made explicit; see [Ray67a], th. 1 (iii) and the proof given below, at the end of section 5.

It follows obviously from (i) that the underlying topological space of Y is the quotient of the underlying topological space of X_0 by the equivalence relation defined by the (Sch/S) -groupoid X_* .

We shall first prove this theorem when X_0 is affine and d_1 is locally free of constant rank n . We shall then see how one can reduce to this special case.

In the case in which we have placed ourselves, X_0, X_1 and X_2 are affine. We can therefore suppose that one has

$$X_i = \text{Spec } A_i, \quad d_j = \text{Spec } \delta_j \quad \text{and} \quad d'_k = \text{Spec } \delta'_k,$$

the A_i being commutative rings and the δ_j, δ'_k ring homomorphisms. One can then replace diagram $(0,1,2)$ by the following

$$((0,1,2)^*) \quad \begin{array}{ccccc} & & \delta'_0 & & \\ & & \swarrow & & \searrow \\ A_2 & \xleftarrow{\delta'_0} & A_1 & \xleftarrow{\delta_0} & A_0 \\ \delta'_2 \uparrow & & \delta'_1 & & \uparrow \delta_1 \\ & & \delta_0 & & \\ A_1 & \xleftarrow{\delta_0} & A_0 & & \\ & & \delta_1 & & \end{array}$$

where the two squares on the left are cocartesian.

Denote by B the subring of A_0 consisting of the $a \in A_0$ such that $\delta_0(a) = \delta_1(a)$.

a) Let us show that A_0 is integral over B . If a belongs to A_0 , let

$$P_{\delta_1}(T, \delta_0(a)) = T^n - \sigma_1 T^{n-1} + \cdots + (-1)^n \sigma_n$$

be the characteristic polynomial of $\delta_0(a)$ when one regards A_1 as an algebra over A_0 by means of the homomorphism δ_1 (cf. Bourbaki, Alg. VIII, §12 and Alg. comm. II, §5, exercise 9). Since the squares on the left of $(0,1,2)^*$ are cocartesian, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_0 \delta_0(a))$$

and

$$\delta_1(P_{\delta_1}(T, \delta_0(a))) = P_{\delta'_2}(T, \delta'_1 \delta_0(a)).$$

Since $\delta'_0 \delta_0 = \delta'_1 \delta_0$, one has

$$\delta_0(P_{\delta_1}(T, \delta_0(a))) = \delta_1(P_{\delta_1}(T, \delta_0(a))),$$

that is, $\delta_0(\sigma_i) = \delta_1(\sigma_i)$ for every i . On the other hand, Hamilton–Cayley tells us that one has

$$\delta_0(a)^n - \delta_1(\sigma_1) \delta_0(a)^{n-1} + \cdots + (-1)^n \delta_1(\sigma_n) = 0.$$

Since $\delta_1(\sigma_i) = \delta_0(\sigma_i)$, one also has

$$\delta_0(a)^n - \delta_0(\sigma_1) \delta_0(a)^{n-1} + \cdots + (-1)^n \delta_0(\sigma_n) = 0,$$

whence

$$a^n - \sigma_1 a^{n-1} + \cdots + (-1)^n \sigma_n = 0,$$

for there exists a homomorphism $\tau : A_1 \rightarrow A_0$ such that $\tau \delta_0 = \text{id}_{A_0}$, so δ_0 is injective. It follows that A_0 is integral over B .

b) Now consider two prime ideals x and y of A_0 . We shall show that the equality $x \cap B = y \cap B$ implies the existence of a prime ideal z of A_1 such that $x = d_0(z)$ and $y = d_1(z)$.

Indeed, if the assertion were not true, x would be distinct from $\delta_0^{-1}(t)$ for every prime ideal t of A_1 such that $\delta_1^{-1}(t) = y$. For such a t one would have $\delta_0^{-1}(t) \cap B = \delta_1^{-1}(t) \cap B = y \cap B = x \cap B$, whence, by Cohen–Seidenberg (cf. Bourbaki, Alg. comm. V, §2, cor. 1 of prop. 1), it would follow that x would not be contained in any $\delta_0^{-1}(t)$.⁽²¹⁾ Now there are at most n prime ideals t of A_1 such that $\delta_1^{-1}(t) = y$ (cf. loc. cit., prop. 3), so, by the “Prime ideal avoidance lemma” (loc. cit., II, §1, prop. 3), there would be an $a \in x$ belonging to none of the $\delta_0^{-1}(t)$.

⁽²¹⁾**N.D.E.** The original has been expanded in what follows; in particular, Lemma 4.1.1, taken from [DG70], III, §2.4, Lemma 4.3, has been added.

Consequently, $\delta_0(a)$ would belong to none of these ideals t , and therefore, by the lemma below, the norm $N_{\delta_1}(\delta_0(a))$ would not belong to y (one calculates this norm by regarding A_1 as an algebra over A_0 by means of the homomorphism δ_1 ; one has $N_{\delta_1}(\delta_0(a)) = \sigma_n$ with the notation of a)). Now, since $(-1)^{n-1}\sigma_n = a^n + \sum_{i=1}^{n-1}(-1)^i\sigma_i a^{n-i}$, this norm belongs to $B \cap x = B \cap y$, whence the contradiction.

Lemma 4.1.1. *Let $A \rightarrow A'$ be a morphism of commutative rings making A' a projective A -module of rank n . Let $\mathfrak{p} \in \operatorname{Spec}(A)$, let $\mathfrak{q}_1, \dots, \mathfrak{q}_r$ be the elements of $\operatorname{Spec}(A')$ above $\mathfrak{p}$, and let $a \in A'$. Then a belongs to $\mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_r$ if and only if its norm $N(a)$ belongs to $\mathfrak{p}$.*

Indeed, replacing A and A' by the localizations $A_{\mathfrak{p}}$ and $A'_{\mathfrak{p}}$, one reduces to the case where $(A, \mathfrak{p})$ is local and A' semilocal, with $\operatorname{Spec}(A') = \{\mathfrak{q}_1, \dots, \mathfrak{q}_r\}$. In this case, A' is a free A -module of rank n (cf. Bourbaki, Alg. comm. II, §3.2, cor. 2 of prop. 5), and $N(a)$ is the determinant of the endomorphism $\ell_a : a' \mapsto aa'$ of A' ; one therefore has the following equivalences:

$$N(a) \notin \mathfrak{p} \iff N(a) \text{ invertible} \iff \ell_a \text{ invertible} \iff a \notin \mathfrak{q}_1 \cup \dots \cup \mathfrak{q}_r.$$

c) Proof of (i):

Put $Y = \operatorname{Spec} B$ and $p = \operatorname{Spec} i$, where i is the inclusion of B into A_0 . By a), the morphism $p : X_0 \rightarrow Y$ is surjective. We shall first show that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces: indeed, it follows from b) that the underlying set of $\operatorname{Spec} B$ is obtained from the underlying set of X_0 by identifying the points x and y such that there exists $z \in X_1$ with $d_1 z = y$, $d_0 z = x$. Moreover, since i is integral, $p = \operatorname{Spec} i$ is closed, so that Y has the quotient topology of that of X_0 . It follows that p is open. Indeed, let U' be any open subset of X_0 ; since d_1 is surjective and finite locally free, hence faithfully flat and of finite presentation, and therefore open, the saturation $U = d_1(d_0^{-1}(U'))$ of U' for the equivalence relation defined by X_* is open. Then $p(U') = p(U)$ is open, since Y has the quotient topology.

Finally, it follows from the choice of B and from the fact that p, d_0 and d_1 are affine that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_{X_0}) \rightrightarrows_{p_*(\delta_0)}^{p_*(\delta_1)} p_*(d_{0*}(\mathcal{O}_{X_1})) = p_*(d_{1*}(\mathcal{O}_{X_1}))$$

is exact.

It remains to show that (Y, p) is also the cokernel of (d_0, d_1) in the category of schemes (more generally in that of spaces ringed in local rings). Let, then, $q : X_0 \rightarrow Z$ be a morphism of schemes such that $qd_0 = qd_1$. By what precedes, there is one and only one morphism of ringed spaces $r : Y \rightarrow Z$ such that $q = rp$. One must show that, for every $y \in Y$, the homomorphism $\mathcal{O}_{r(y)} \rightarrow \mathcal{O}_y$ induced by r is local. This follows from the fact that p is surjective, hence y has the form $p(x)$, and that the homomorphism $\mathcal{O}_{q(x)} \rightarrow \mathcal{O}_x$ induced by q is local.

d) Proof of (ii): follows from a) and c).

e) Proof of (iii):

Recall that $\underline{P}$ denotes the underlying set of a scheme P , and $\underline{d} : \underline{P} \rightarrow \underline{Q}$ the map induced by a morphism $d : P \rightarrow Q$.

Lemma 4.1.2. ⁽²²⁾ *Let $(A, \mathfrak{m})$ be a local ring, k its residue field, and K an extension of the field k . Then there exists a local flat A -algebra B such that $B/\mathfrak{m}B$ is k -isomorphic to K ; moreover, one can choose B finite and free over A if K has finite degree over k .*

This is proved in EGA 0_{III}, 10.3.1, where it is further shown that one can choose B noetherian if A is. For the reader's convenience, let us indicate the proof.

⁽²²⁾**N.D.E.** This lemma, used several times in this exposé and in subsequent exposés (VIA, VIB), has been inserted. It appeared as Lemma VIB, 4.5.1 in the original edition of SGAD (1965).

Put $A' = A[T]$, where T is an indeterminate. If $K = k(T)$, let $\mathfrak{p} = \mathfrak{m}A'$ and $B = A'_{\mathfrak{p}}$. Then $B/\mathfrak{m}B \simeq k[T]_{(0)} = k(T)$, and B is flat over A' , which is a free A -module, so B is flat over A .

If $K = k(t) = k[t]$, where t is algebraic over k , put $B = A'/(F)$, where $F \in A'$ is a monic polynomial whose image in $k[T]$ is the minimal polynomial f of t over k . Then B is a free A -module of finite rank $\deg(F) = \deg(f)$. In particular, B is integral over A , so every maximal ideal of B contains $\mathfrak{m}$. Since $B/\mathfrak{m}B \simeq k[T]/(f) \simeq K$, it follows that B is local with maximal ideal $\mathfrak{m}B$. This already shows that if $[K : k] < \infty$, one can choose B finite and free over A .

In the general case, let $(t_i)_{i \in I}$ be a system of generators of K over k , and give I a well-order (that is, a total order $\leq$ such that every nonempty subset of I has a least element). For every $i \in I$, denote by k_i (resp. $k_{<i}$) the subfield of K generated by the t_j for $j \leq i$ (resp. $j < i$). By adding an element if necessary, one can suppose that I has a greatest element ξ , so that $K = k_{\xi}$. Consider the subset J of I consisting of those indices i for which there exists an inductive system $(A_j)_{j \leq i}$ of local flat A -algebras such that $A_j/\mathfrak{m}A_j \simeq k_j$ and A_j is flat over A_{ℓ} for every $\ell < j$. Suppose $I - J$ nonempty; let i be its least element and let $A' = \varinjlim_{j < i} A_j$. Since tensor product commutes with inductive limits, A' is flat over A and over each A_j for $j < i$, and one has $A'/\mathfrak{m}A' \simeq A' \otimes_A (A/\mathfrak{m}) \simeq k_{<i}$. Moreover, A' is local with maximal ideal $\mathfrak{m}A'$. Indeed, if $x = f_j(x_j)$ is not invertible, then x_j is not invertible, hence belongs to the maximal ideal $\mathfrak{m}A_j$ of A_j , whence $x \in \mathfrak{m}A'$. It then follows from the monogenic case treated above that there exists a local flat A' -algebra A_i such that $A_i/\mathfrak{m}A_i \simeq k_{<i}(t_i) = k_i$; then A_i is flat over each A_j for $j < i$, and hence $i \in J$, contrary to the hypothesis. This contradiction shows that $J = I$, and therefore A_{ξ} answers the question. Lemma 4.1.2 is proved.

Let us now prove 4.1 (iii). Recall that $\underline{P}$ denotes the underlying set of a scheme P , and $\underline{d} : \underline{P} \rightarrow \underline{Q}$ the map induced by a morphism $d : P \rightarrow Q$. One can then express b) by saying that the map

$$d_0 \boxtimes d_1 : \underline{X}_1 \rightarrow \underline{X}_0 \times_Y \underline{X}_0$$

with components $\underline{d}_0$ and $\underline{d}_1$ is surjective; now this map factors as follows

$$\underline{X}_1 \xrightarrow{d_0 \boxtimes d_1} \underline{X}_0 \times_Y \underline{X}_0 \xrightarrow{q} \underline{X}_0 \times_Y \underline{X}_0,$$

q being the canonical map; the image of $d_0 \boxtimes d_1$ therefore contains all points v of $X_0 \times_Y X_0$ such that $\{v\} = q^{-1}(q(v))$. This last condition⁽²³⁾ holds, in particular, if v is rational over Y , that is, if the residue field $\kappa(v)$ of v is identified with the residue field $\kappa(w)$ of the image w of v in Y .

If $v \in X_0 \times_Y X_0$ is not rational over Y , let w again be its image in Y . By Lemma 4.1.2, there exists a local ring C with radical $\mathfrak{m}$ and a local flat homomorphism $f : \mathcal{O}_w \rightarrow C$ such that $C/\mathfrak{m}$ is isomorphic to $\kappa(v)$ as a $\kappa(w)$ -algebra. If one puts $Y' = \text{Spec } C$ and denotes by $\pi : Y' \rightarrow Y$ the morphism induced by f , it is clear that the canonical projection from $(X_0 \times_Y X_0) \times_Y Y'$ to $X_0 \times_Y X_0$ sends to v a point v' of $(X_0 \times_Y X_0) \times_Y Y'$ that is rational over Y' . Since

$$(X_0 \times_Y X_0) \times_Y Y' \simeq (X_0 \times_Y Y') \times_{Y'} (X_0 \times_Y Y'),$$

and since the hypotheses of Theorem 4.1 and the preceding results, in particular point b), remain valid after the base change $\pi : Y' \rightarrow Y$, v' is the image of an element $u' \in X_1 \times_Y Y'$ under the morphism obtained from $d_0 \boxtimes d_1$ by base change. If u is the image of u' in X_1 , one indeed has $v = (d_0 \boxtimes d_1)(u)$.

f) Proof of (iv):⁽²⁴⁾

⁽²³⁾**N.D.E.** Note the permutation of the pages in Lecture Notes 151: the actual order is 265–266–268–269–267–270–271.

⁽²⁴⁾**N.D.E.** The following lemma, taken from [DG70], I, §5, Prop. 1.5 (see also the proof of EGA IV₃, 8.11.5), has been added; it is used implicitly in the original and explicitly in [DG70], III, §2, 4.6. It is also useful in th. 7.1 below.

Lemma 4.1.3. *If a monomorphism of schemes $f : T \rightarrow Z$ is a finite morphism, it is a closed immersion.*

Indeed, by covering Z with affine open subsets Z_i and replacing f by the induced morphisms $f^{-1}(Z_i) \rightarrow Z_i$, one reduces (f being finite, hence affine) to the case $Z = \operatorname{Spec} B$ and $T = \operatorname{Spec} A$. Since f is a monomorphism, the diagonal morphism $T \rightarrow T \times_Z T$ is an isomorphism (EGA I, 5.3.8), i.e. $A \otimes_B A \rightarrow A$ is an isomorphism. Consequently, for every maximal ideal $\mathfrak{m}$ of B , one has an isomorphism

$$(A/\mathfrak{m}A) \otimes_k (A/\mathfrak{m}A) \simeq (A/\mathfrak{m}A),$$

where one has put $k = B/\mathfrak{m}$. Since A is finite over B , $A/\mathfrak{m}A$ is a finite-dimensional k -vector space of dimension d , and the isomorphism above implies that $d = 0$ or 1 , so that the morphism $k = B/\mathfrak{m} \rightarrow A/\mathfrak{m}A$ is surjective. Thus, by Nakayama's lemma (A being finite over B), the morphism $B_{\mathfrak{m}} \rightarrow A_{\mathfrak{m}}$ is surjective. It follows that the morphism of B -modules $B \rightarrow A$ is surjective (since its cokernel C satisfies $C_{\mathfrak{m}} = 0$ for every $\mathfrak{m}$, and is therefore zero). This proves the lemma.

Let us now prove (iv). By hypothesis $X_0 = \operatorname{Spec} A_0$, $X_1 = \operatorname{Spec} A_1$, and, for $i = 0, 1$, the morphism $\delta_i : A_0 \rightarrow A_1$ makes A_1 a finitely generated A_0 -module; thus, a fortiori, the morphism $A_0 \otimes_B A_0 \rightarrow A_1$ is finite.

One further supposes that $d = d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times_Y X_0$ is a monomorphism; thus, by the preceding lemma, the morphism $A_0 \otimes_B A_0 \rightarrow A_1$ is surjective.

We shall show that it is an isomorphism (in the course of the proof we shall show that $p : X_0 \rightarrow Y$ is finite locally free). It suffices to show that, for every prime ideal $\mathfrak{p}$ of B , the homomorphism $(A_0)_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} (A_0)_{\mathfrak{p}} \rightarrow (A_1)_{\mathfrak{p}}$ with components $\delta_{0\mathfrak{p}}$ and $\delta_{1\mathfrak{p}}$ is bijective. In other words, one may suppose B local. It then follows from b) that $(A_0)_{\mathfrak{p}}$ is semilocal; indeed, if $\mathfrak{m}$ is a maximal ideal of $(A_0)_{\mathfrak{p}}$, the other maximal ideals have the form $\delta_0^{-1}(\mathfrak{n})$, where $\mathfrak{n}$ ranges over the prime ideals of A_1 such that $\delta_1^{-1}(\mathfrak{n}) = \mathfrak{m}$; the assertion therefore follows from the fact that there are at most $n = [A_1 : A_0]$ such prime ideals $\mathfrak{n}$. By making a faithfully flat base change if necessary,⁽²⁵⁾ one can also suppose that the residue field of B is infinite, so that one can use the following lemma:

Lemma 4.2. *Let B be a local ring with infinite residue field, A a semilocal ring, and $i : B \rightarrow A$ a homomorphism sending the maximal ideal $\mathfrak{n}$ of B into the radical $\mathfrak{r}$ of A . Let M be a free A -module of rank n and N a B -submodule of M generating M as an A -module. Then N contains a basis of M over A .*

Indeed, recall that a sequence $m_1, \dots, m_n$ of elements of M is an A -basis of M if and only if the canonical images of $m_1, \dots, m_n$ in $M/\mathfrak{r}M$ form a basis of $M/\mathfrak{r}M$ over $A/\mathfrak{r}$. One can therefore replace M by $M/\mathfrak{r}M$, N by $N/(N \cap \mathfrak{r}M)$, A by $A/\mathfrak{r}$ and B by $B/\mathfrak{m}$. In this case the lemma is easy (if A is a product of fields $K_1 \times \dots \times K_r$, one can identify M with the module $K_1^n \times \dots \times K_r^n$; if x_j is an element of N whose j th component in $K_1^n \times \dots \times K_r^n$ is nonzero, show that a certain linear combination x of the x_j with coefficients in B has all its components nonzero; then replace M by M/Ax and proceed by induction on n).

We apply the preceding lemma in the following situation: $B = B$, $A = A_0$, i is the inclusion of B into A_0 , $M = A_1$ regarded as an A_0 -module by means of the homomorphism δ_1 , $N = \delta_0(A_0)$. Indeed, since $d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times_Y X_0$ is a closed immersion, the homomorphism $A_0 \otimes_B A_0 \rightarrow A_1$ with components δ_0 and δ_1 is surjective; this means precisely that $\delta_0(A_0)$ generates the A_0 -module A_1 .

Let, then, $a_1, \dots, a_n$ be elements of A_0 such that $\delta_0(a_1), \dots, \delta_0(a_n)$ form a basis of A_1 over A_0 . If we show that $a_1, \dots, a_n$ is a basis of A_0 over B , it will follow that the homomorphism $A_0 \otimes_B A_0 \rightarrow A_1$ sends the basis $(1 \otimes a_i)_{1 \leq i \leq n}$ to the basis $(\delta_0(a_i))_{1 \leq i \leq n}$, and is therefore bijective. Consequently, if $\varepsilon : \mathbf{Z}^n \rightarrow A_0$ is the homomorphism of abelian groups sending

⁽²⁵⁾**N.D.E.** Cf. Lemma 4.1.2.

the natural basis of $\mathbf{Z}^n$ to $a_1, \dots, a_n$, it suffices to prove that the map $B \otimes_{\mathbf{Z}} \mathbf{Z}^n \rightarrow A_0$ with components i and ε is bijective.

Now diagram $(0, 1, 2)^*$ considered at the beginning of this proof induces the following commutative diagram:

$$\begin{array}{ccccc} A_2 & \xleftarrow{\delta'_0} & A_1 & \xleftarrow{\delta_0} & A_0 \\ \uparrow u_2 & & \uparrow u_1 \simeq & & \uparrow u_0 \\ A_1 \otimes_{\mathbf{Z}} \mathbf{Z}^n & \xleftarrow[\delta_1 \otimes \mathbf{Z}^n]{\delta_0 \otimes \mathbf{Z}^n} & A_0 \otimes_{\mathbf{Z}} \mathbf{Z}^n & \xleftarrow{i \otimes \mathbf{Z}^n} & B \otimes_{\mathbf{Z}} \mathbf{Z}^n, \end{array}$$

where u_0, u_1 and u_2 have components respectively i and ε , δ_1 and $\delta_0\varepsilon$, δ'_2 and $\delta'_0\delta_0\varepsilon$. We know that u_1 is an isomorphism. Since the two squares on the left of $(0, 1, 2)^*$ are cocartesian, u_2 is bijective. Now the two horizontal rows of our diagram are exact; thus u_0 is bijective.⁽²⁶⁾ This shows that A_0 is a locally free B -module of rank n , and, by the preceding reductions, this implies that $\delta_0 \otimes \delta_1 : A_0 \otimes_B A_0 \rightarrow A_1$ is an isomorphism. This completes the proof of Theorem 4.1 in the special case considered (X_0 affine and d_1 locally free of constant rank n).

5. Passage to the quotient by a finite flat groupoid (general case)

a) Let $U^{(n)}$ be the largest open subset of X_0 above which d_1 is finite locally free of rank n . One knows that X_0 is the direct sum of the $U^{(n)}$. It follows, on the other hand, from the two cartesian squares

$$\begin{array}{ccc} X_2 & \xrightarrow{d'_0} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_0} & X_0 \end{array} \quad \text{and} \quad \begin{array}{ccc} X_2 & \xrightarrow{d'_1} & X_1 \\ d'_2 \downarrow & & \downarrow d_1 \\ X_1 & \xrightarrow{d_1} & X_0 \end{array}$$

that the inverse images of $U^{(n)}$ under d_0 and d_1 both coincide with the largest open subset of X_1 above which d'_2 is locally free of rank n ;⁽²⁷⁾ one therefore has $d_0^{-1}(U^{(n)}) = d_1^{-1}(U^{(n)})$, so that the groupoid X_* is the direct sum of the groupoids $X_*^{(n)}$ induced by X_* on the open and closed subsets $U^{(n)}$. Consequently, as one easily sees, it suffices to prove Theorem 4.1 for each $X_*^{(n)}$: one is reduced to the case where d_1 is finite locally free of rank n .

b) We are now in a position to prove our theorem in the general case.

By a), one can suppose d_1 locally free of rank n . Let, then, (Y, p) be a cokernel of (d_0, d_1) in the category of all ringed spaces. The argument at the end of paragraph 4.c) shows that, to prove 4.1 (i), it suffices to prove that Y is a scheme and $p : X_0 \rightarrow Y$ a morphism of schemes. By Lemma 1.2, the question is local on Y : let $y \in Y$ and $x \in X_0$ satisfy $p(x) = y$; if x has an affine saturated open neighborhood U , $p(U)$ will be an affine open subset of Y by §4, and $p|_U$ will be a morphism of schemes. It therefore suffices to prove that every $x \in X_0$ has an affine saturated open neighborhood U . Here is how one proceeds (the proof is taken from SGA 1, VIII, cor. 7.6).

$$\begin{array}{ccccccccccc} d_1(d_0^{-1}(x)) \subset & U = (V_f)' & \subset & V_f & \subset & V' & \subset & V & \subset & X_0 \\ & \uparrow & & \uparrow & & \uparrow & & \uparrow & & \\ & \text{largest saturated} & & \text{distinguished affine} & & \text{largest saturated} & & \text{affine open subset} & & \\ & \text{open subset of } V_f & & \text{open subset of } V & & \text{open subset of } V & & & & \end{array}$$

⁽²⁶⁾N.D.E. What follows has been added.

⁽²⁷⁾N.D.E. Indeed, since d_0 (resp. d_1) is surjective, flat and finite, hence faithfully flat and affine, d'_2 has rank n above a neighborhood of a point x of X_1 if and only if d_1 has rank n above a neighborhood of $d_0(x)$ (resp. $d_1(x)$).

By condition b) of 4.1, there exists an affine open subset V of X_0 containing $d_1(d_0^{-1}(x))$; ⁽²⁸⁾ if $F = X_0 - V$, $d_1(d_0^{-1}(F))$ is closed because d_1 is integral, and $V' = X_0 - d_1(d_0^{-1}(F))$ is the largest saturated open subset contained in V . Since V' is a neighborhood of the finite set $d_1(d_0^{-1}(x))$, there exists a section f of the structure sheaf of V which vanishes on $V - V'$ and such that $d_1(d_0^{-1}(x))$ is contained in the open subset V_f of V consisting of the points where f does not vanish. We shall see that the largest saturated open subset $(V_f)'$ of V_f is affine, and thus answers the question.

Indeed, let $Z(f) = V' - V_f$. Then $d_0^{-1}(Z(f))$ is the set of points of $d_0^{-1}(V') = d_1^{-1}(V')$ where the image $d_0^*(f)$ of f under the map induced by d_0 vanishes. On the other hand, since d_1 induces a locally free morphism of rank n from $d_0^{-1}(V') = d_1^{-1}(V')$ onto V' , ⁽²⁹⁾ by Lemma 4.1.1, $d_1(d_0^{-1}(Z(f)))$ is the set of points where the norm N of $d_0^*(f)$ for the morphism d_1 vanishes. It follows that $(V_f)' = V' - d_1(d_0^{-1}(Z(f)))$ is the set of points of V_f where N does not vanish; consequently $(V_f)'$ is affine.

This proves 4.1 (i); assertions (ii), (iii), and the first part of (iv) are then clear. Finally, let us show the consequences noted at the end of point (iv) (cf. [Ray67a], th. 1 (iii)).

By hypothesis, the groupoid X_* comes from an equivalence relation $i : R \rightarrow X_0 \times X_0$ (i thus being an immersion, cf. N.D.E. (19)), and one has established that R is effective (cf. Exp. IV, 3.3.2) and that $p : X_0 \rightarrow Y = X_0/R$ is a surjective finite locally free morphism, hence, in particular, faithfully flat and of finite presentation. Consequently, denoting by (M) the family of faithfully flat morphisms locally of finite presentation, R is (M) -effective. Thus, by Exp. IV, 6.3.3, (Y, p) represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from IV, 3.4.3.1.

Remark 5.1. ⁽³⁰⁾ Retain the hypotheses and notation of 4.1, and further suppose S locally noetherian and $\pi_0 : X_0 \rightarrow S$ quasi-projective. Let us then show that $\pi : Y \rightarrow S$ is quasi-projective.

The hypotheses above imply that $Y \rightarrow S$ is of finite type; see the proof of 6.1 (ii). Let $\mathcal{A}$ be an invertible $\mathcal{O}_{X_0}$ -module ample for π_0 . By EGA II, 6.1.12, $p_*(\mathcal{A})$ is an invertible $p_*(\mathcal{O}_{X_0})$ -module. There therefore exists a covering $(V_i)_{i \in I}$ of Y by affine open subsets such that $\mathcal{A}$ is trivial above each of the saturated affine open subsets $U_i = p^{-1}(V_i)$.

For each index i , denote by $A_{i,0} = \mathcal{O}_{X_0}(U_i)$, by $A_{i,1}$ the ring of the affine open subset $d_0^{-1}(U_i) = d_1^{-1}(U_i)$ of X_1 , by $\delta_{i,0}$ (resp. $\delta_{i,1}$) the morphism $A_{i,0} \rightarrow A_{i,1}$ induced by d_0 (resp. d_1), and by $B_i = \mathcal{O}_Y(V_i) = \{b \in A_{i,0} \mid \delta_{i,0}(b) = \delta_{i,1}(b)\}$.

Following EGA II, §6.5, consider the invertible $\mathcal{O}_{X_0}$ -module $N_{d_1}(d_0^*(\mathcal{A}))$, the norm, relative to the finite locally free morphism $d_1 : X_1 \rightarrow X_0$, of the invertible $\mathcal{O}_{X_1}$ -module $d_0^*(\mathcal{A})$. If $\mathcal{A}$ is given, relative to the covering $(U_i)_{i \in I}$, by transition functions $c_{ij} \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times$, then $N_{\delta_1}(d_0^*(\mathcal{A}))$ is given by the transition functions $N_{d_1}(\delta_0(c_{ij})) \in \mathcal{O}_{X_0}(U_i \cap U_j)^\times$; since, by paragraph 4.a), these elements belong to $\mathcal{O}_Y(V_i \cap V_j)^\times$, they define an invertible $\mathcal{O}_Y$ -module $\mathcal{L}$ such that $p^*(\mathcal{L}) = N_{d_1}(d_0^*(\mathcal{A}))$. Note further that, for every $n \in \mathbf{N}^*$, one has $p^*(\mathcal{L}^n) = N_{d_1}(d_0^*(\mathcal{A}^n))$, cf. loc. cit., (6.5.2.1).

Let us show that $\mathcal{L}$ is ample for the morphism $\pi : Y \rightarrow S$. To this end, replacing S by an affine open subset, one can suppose S affine. Let, then, $y \in Y$, $x \in X_0$ such that $p(x) = y$, V an affine open subset of Y containing y , and $U = p^{-1}(V)$. Since $\mathcal{A}$ is π_0 -ample, there exist $n \in \mathbf{N}^*$ and a section $s \in \Gamma(X_0, \mathcal{A}^n)$ such that the open subset $(X_0)_s$ satisfies $x \in (X_0)_s \subset U$. With the preceding notation, s is given by sections $a_i \in A_{i,0} = \mathcal{O}_{X_0}(U_i)$ such that $a_i = c_{ij}a_j$ on $U_i \cap U_j$, and $(X_0)_s$ is the union of the open subsets $U'_i = \{\mathfrak{p} \in \text{Spec}(A_{i,0}) \mid a_i \notin \mathfrak{p}\}$.

For each index i , put $N(a_i) = N_{\delta_1}(\delta_0(a_i)) \in B_i$. By 4.1 (i) and Lemma 4.1.1, one has:

$$p(U'_i) = p d_1 d_0^{-1}(U'_i) = p d_1(\{\mathfrak{q} \in \text{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\})$$

⁽²⁸⁾ N.D.E. One has $d_1(d_0^{-1}(x)) = d_0(d_1^{-1}(x))$, cf. N.D.E. (16) in Theorem 4.1.

⁽²⁹⁾ N.D.E. The reference to Lemma 4.1.1 has been added, cf. [DG70], III, §5.2, p. 313.

⁽³⁰⁾ N.D.E. This paragraph has been added.

and $d_1(\{\mathfrak{q} \in \operatorname{Spec}(A_{i,1}) \mid \delta_{i,0}(a_i) \notin \mathfrak{q}\}) = \{\mathfrak{p} \in \operatorname{Spec}(A_{i,0}) \mid N_{\delta_1}(\delta_{i,0}(a_i)) \notin \mathfrak{p}\}$, whence

$$p(U'_i) = \{\mathfrak{p} \in \operatorname{Spec}(B_i) \mid N(a_i) \notin \mathfrak{p}\}.$$

It follows that $p((X_0)_s)$ equals $Y_{N(s)}$, where $N(s)$ denotes the section of $\mathcal{L}^n$ on Y defined by the sections $N(a_i) \in \mathcal{O}_Y(V_i)$. One therefore has

$$(*) \quad y \in p((X_0)_s) = Y_{N(s)} \subset p(U) = V.$$

This shows that $\mathcal{L}$ is ample for $\pi : Y \rightarrow S$, which completes the proof that $\pi : Y \rightarrow S$ is quasi-projective.

6. Passage to the quotient when a quasi-section exists

We shall now prove a technical lemma which will serve us in the proof of the two theorems we have in view. Let S be a scheme and

$$X_2 \xrightarrow[d'_0, d'_1, d'_2]{d_0, d_1} X_1 \xrightarrow[d_0, d_1]{d_0, d_1} X_0$$

a (Sch/ S)-groupoid. We shall call a quasi-section of the groupoid X_* any subscheme U of X_0 such that (1) and (2) hold:

(1) The restriction v of d_1 to $d_0^{-1}(U)$ is a finite, locally free and surjective morphism from $d_0^{-1}(U)$ onto X_0 .

(2) Every subset E of U consisting of points pairwise equivalent for the equivalence relation defined by X_* (§3.e)) is contained in an affine open subset of U .⁽³¹⁾

If U is a quasi-section of X_* , the (Sch/ S)-groupoid

$$U_2 \xrightarrow[u'_0, u'_1, u'_2]{u_0, u_1} U_1 \xrightarrow[u_0, u_1]{u_0, u_1} U$$

induced by X_* and the inclusion of U into X_0 (cf. §3.a)) satisfies the hypotheses of Theorem 4.1. Indeed, put $V = d_0^{-1}(U)$ and let u and v be the morphisms with source V induced respectively by d_0 and d_1 :

$$X_0 \xleftarrow{v} V \xrightarrow{u} U.$$

By paragraph 3.b), one has a cartesian square

$$\begin{array}{ccc} U_1 & \longrightarrow & V \\ u_1 \downarrow & & \downarrow v \\ U & \xrightarrow{\text{inclusion}} & X_0, \end{array}$$

so u_1 is surjective and finite locally free by (1). Together with (2), condition (1) therefore ensures that the groupoid U_* satisfies the hypotheses of Theorem 4.1. In particular, $\operatorname{Coker}(u_0, u_1)$ exists in (Sch/ S). Moreover, d_0 has a section, so u is an effective universal epimorphism (cf. III 1.12); it follows, by Proposition 3.1, that $\operatorname{Coker}(u_0, u_1)$ coincides with the cokernel $\operatorname{Coker}(v_0, v_1)$ of the groupoid V_* :

$$V_2 \xrightarrow[v'_0]{v'_1} V_1 \xrightarrow[v_0]{v_1} V,$$

the inverse image of U_* by the base change $u : V \rightarrow U$, that is, also the inverse image of X_* by the base change:

$$V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_0} X_0.$$

⁽³¹⁾**N.D.E.** If $x, y \in E$, there exists $z \in X_1$ such that $d_1(x) = z$ and $d_0(z) = y$, that is, z belongs to the set $v^{-1}(x)$, which is finite by (1). Thus E is contained in the finite set $d_0(v^{-1}(x)) = d_0 d_1^{-1}(x) \cap U$.

(32) By paragraph 3.c), V_* is isomorphic to the groupoid V'_* , the inverse image of X_* by the base change:

$$v : V \xrightarrow{\text{inclusion}} X_1 \xrightarrow{d_1} X_0,$$

and hence V'_* has a cokernel in (Sch/S) . Now, being flat, surjective and finite, $v : V \rightarrow X_0$ is faithfully flat and quasi-compact, hence an effective universal epimorphism by III 6.3.2. Consequently, by Proposition 3.1, the groupoid X_* has a cokernel $\text{Coker}(d_0, d_1)$ in (Sch/S) . We have therefore proved the first assertion of point (i) of the following lemma:

Lemma 6.1. *Suppose the (Sch/S) -groupoid X_* has a quasi-section. Then:*

- (i) *There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S) ; moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*
- (i') *p is surjective, and is open (resp. universally closed) if d_0 is.*
- (ii) *Suppose S locally noetherian and X_0 locally of finite type (resp. of finite type) over S . Then p and $Y \rightarrow S$ are locally of finite presentation (resp. of finite presentation).*
- (iii) *The morphism $X_1 \rightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (iv) *If (d_0, d_1) is an equivalence pair, $X_1 \rightarrow X_0 \times_Y X_0$ is an isomorphism. Moreover, if $d_0 : X_1 \rightarrow X_0$ is flat, p is faithfully flat.*

Before proving the second assertion of (i), we shall prove (i'), (ii) and (iii).

a) Proof of (i') and (ii):

We have just seen that (Y, p) is identified with $\text{Coker}(v_0, v_1)$ and $\text{Coker}(u_0, u_1)$. Let, then, q and r be the canonical epimorphisms from U and V to Y :

$$\begin{array}{ccccc} X_0 & \xleftarrow{v} & V & \xrightarrow{u} & U \\ & \searrow p & \downarrow r & \swarrow q & \\ & & Y & & \end{array}$$

By hypothesis, v is surjective and finite locally free, hence open. On the other hand, if $d_0 : X_1 \rightarrow X_0$ is open (resp. universally closed), then u , obtained from it by base change, is also so. Since, by Theorem 4.1, q is surjective, integral and open, r is surjective, and is open (resp. universally closed) if d_0 is. The same therefore holds for p , since v is surjective. This proves (i').

Now suppose S locally noetherian and X_0 locally of finite type over S , so that X_0 is locally noetherian.

Let us show that Y is locally of finite presentation over S . Let $S' = \text{Spec } R$ be an affine open subset of S , $Y' = \text{Spec } B$ an affine open subset of Y projecting into S' , and $U' = \text{Spec } A$ the inverse image of Y' in U . Since R is noetherian, it suffices to show that B is a finitely generated R -algebra; now, by paragraphs 4 and 5, B is contained in A , which is a finitely generated R -algebra; the assertion therefore follows from the facts that R is noetherian and A integral over B .

Finally, since $X_0 \rightarrow S$ is locally of finite type, p is also so (EGA I, 6.6.6), hence p is locally of finite presentation since Y is locally noetherian.

It remains to show the last assertion of (ii). Further suppose X_0 of finite type over S . Then, since p is surjective, Y is moreover quasi-compact over S , hence of finite type over S . Since S is locally noetherian, $X_0 \rightarrow S$ and $Y \rightarrow S$ are of finite presentation, and hence $p : X_0 \rightarrow Y$ is also so (EGA IV₁, 1.6.2 (v)).

b) Proof of (iii):

Since the groupoid V_* with base V is isomorphic both to the inverse image of U_* by the base change u and to the inverse image of X_* by the base change v , one has a double

(32) **N.D.E.** What follows has been slightly modified; in particular, in Lemma 6.1, the additional hypothesis that d_0 is flat has been moved to (iv), (ii) has been split into (i') and (ii), and the second assertion of (i') has been added.

cartesian square

$$\begin{array}{ccccc} X_1 & \xleftarrow{\quad} & V_1 & \xrightarrow{\quad} & U_1 \\ d_0 \boxtimes d_1 \downarrow & & v_0 \boxtimes v_1 \downarrow & & u_0 \boxtimes u_1 \downarrow \\ X_0 \times_Y X_0 & \xleftarrow[v \times v]{} & V \times_Y V & \xrightarrow[u \times u]{} & U \times_Y U. \end{array}$$

Since $u_0 \boxtimes u_1$ is surjective, so is $v_0 \boxtimes v_1$. Since $v \times v$ is surjective, so is the composite morphism $V_1 \longrightarrow X_0 \times_Y X_0$, and therefore $d_0 \boxtimes d_1$.

c) Proof of (i):

It remains to prove that (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces. We first show that Y is obtained from X_0 by identifying points x and y such that there exists $z \in X_1$ with $d_0(z) = x$ and $d_1(z) = y$. Indeed, p is surjective and one has $pd_0 = pd_1$; moreover, if $p(x) = p(y)$, there is a point z' of $X_0 \times_Y X_0$ whose first projection is x and second projection y . If z is a point of X_1 such that $(d_0 \boxtimes d_1)(z) = z'$, one indeed has $d_0(z) = x$ and $d_1(z) = y$.

On the other hand, if W is a saturated open subset of X_0 , $W \cap U$ is a saturated open subset of U ; by 4.1, $q(W \cap U)$ is an open subset of Y . Since $q(W \cap U)$ is none other than $p(W)$, one sees that Y has the quotient topology of that of X_0 .

It remains to show that the canonical sequence of sheaves of rings

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_{X_0}) \rightrightarrows p_*d_{0*}(\mathcal{O}_{X_1}) = p_*d_{1*}(\mathcal{O}_{X_1})$$

is exact.

Let, then, Y' be an open subset of Y , and put $U' = q^{-1}(Y')$, $X'_0 = p^{-1}(Y')$, etc.⁽³³⁾ Then U' is an open subset of U , saturated for the equivalence relation defined by the groupoid U_* , and it follows from Lemmas 1.1 and 1.2 that Y' is the cokernel, in (Sch/S) and in (Esp. An) , of the groupoid induced by U_* on U' . Similarly, X'_0 is an open subset of X_0 , saturated for the equivalence relation defined by X_* , and one has the following commutative diagram, whose two squares are cartesian:

$$\begin{array}{ccccc} X'_0 & \xleftarrow{\tilde{d}_1} & V' = d_0^{-1}(U') & \xrightarrow{\tilde{d}_0} & U' \\ \downarrow & & \downarrow & & \downarrow \\ X_0 & \xleftarrow{d_1} & V = d_0^{-1}(U) & \xrightarrow{d_0} & U. \end{array}$$

Then $\tilde{d}_1$ is surjective and finite locally free. On the other hand, let $x \in U'$. Since U is a quasi-section, the set $E := d_0 d_1^{-1}(x) \cap U$ is finite and contained in an affine open subset W of U . Then $E' = E \cap U'$ is a finite set contained in the quasi-affine open subset $W \cap U'$. Consequently, there exists an affine open subset W' of $W \cap U'$ containing E' . This shows that U' is a quasi-section of the groupoid X'_* induced by X_* on X'_0 . The first assertion of (i), applied to X'_* and U' , then shows that Y' is the cokernel in (Sch/S) of X'_* .

In particular, for every S -scheme T , one has the exact sequence

$$T(Y') \xrightarrow{T(p|_{X'_0})} T(X'_0) \xrightarrow[T(d_0|_{X'_1})]{T(d_1|_{X'_1})} T(X'_1).$$

Now, if T is the “affine line” $\mathbf{G}_{a,S}$ (I 4.3), this sequence is identified with the sequence:

$$\Gamma(Y', \mathcal{O}_Y) \rightarrow \Gamma(p^{-1}(Y'), \mathcal{O}_{X_0}) \xrightarrow[\delta_0]{\delta_1} \Gamma(d_0^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}) = \Gamma(d_1^{-1}p^{-1}(Y'), \mathcal{O}_{X_1}),$$

which is therefore exact for every open subset Y' . This completes the proof of 6.1 (i).

⁽³³⁾**N.D.E.** What follows has been expanded, in particular the fact that U' is a quasi-section of the groupoid induced on X'_0 .

d) Proof of (iv):

If (d_0, d_1) is an equivalence pair, so is (u_0, u_1) . It follows that $u_0 \boxtimes u_1 : U_1 \rightarrow U \times_Y U$ is an isomorphism (Theorem 4.1), and therefore also $v_0 \boxtimes v_1$ (cf. the cartesian squares of b)); since $v \times v$ is faithfully flat and quasi-compact, $d_0 \boxtimes d_1$ is an isomorphism (SGA 1, VIII 5.4).

Moreover, if d_0 is flat, so is u . Now q is flat by Theorem 4.1, hence so is r . Since v is faithfully flat, p is flat, and therefore faithfully flat since it is surjective.

7. Quotient by a proper flat groupoid

Theorem 7.1. ⁽³⁴⁾ *Let S be a locally noetherian scheme and*

$$X_2 \begin{array}{c} \xrightarrow{d_1'} \\ \xrightarrow{d_0'} \end{array} X_1 \xrightarrow{d_1} X_0$$

a (Sch/S) -groupoid such that d_1 is proper and flat, X_0 is quasi-projective over S ,⁽³⁵⁾ and the morphism $d : X_1 \rightarrow X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite. Then:

- (i) *There exists a cokernel (Y, p) of (d_0, d_1) in (Sch/S) ; moreover, such a (Y, p) is a cokernel of (d_0, d_1) in the category of all ringed spaces.*
- (ii) *p is surjective, open, proper, of finite presentation, and $Y \rightarrow S$ is of finite presentation and separated.*⁽³⁶⁾
- (iii) *The morphism $X_1 \rightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (iv) *If (d_0, d_1) is an equivalence pair, $X_1 \rightarrow X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.*⁽³⁷⁾ *Moreover, (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology and, for every base change $Y' \rightarrow Y$, Y' is the cokernel of the groupoid $X_* \times_Y Y'$ obtained from X_* by the base change $X_0 \times_Y Y' \rightarrow X_0$.*

In particular, for every base change $S' \rightarrow S$, $Y' = Y \times_S S'$ is the cokernel of the S' -groupoid $X'_ = X_* \times_S S'$. Thus, in this case, “formation of the quotient commutes with base change”.*

Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Lemma 1.2 shows that, to prove (i), it suffices to show that every point z of X_0 has a saturated open neighborhood U_z such that, denoting by $\tilde{d}_0$ and $\tilde{d}_1$ the restrictions of d_0 and d_1 to $d_0^{-1}(U_z) = d_1^{-1}(U_z)$, and by (Q, q) the cokernel of $(\tilde{d}_0, \tilde{d}_1)$ in (Esp.An) , Q is a scheme and q a morphism of schemes.

By Lemma 6.1 (i), it therefore suffices to show that every point z of X_0 has a saturated open neighborhood U_z such that the groupoid induced on U_z by X_* has a quasi-section. One can even suppose that z is closed in the fiber of z over S (we shall say that z is closed relative to S).⁽³⁸⁾ The existence of U_z then follows from the lemmas below:

⁽³⁴⁾**N.D.E.** Let us mention here the article of S. Keel and S. Mori ([KM97]), in which the following theorem is established. Let X be an algebraic space of finite type over a locally noetherian base S and $j : R \rightarrow X \times_S X$ a flat groupoid whose stabilizer $j^{-1}(\Delta_X)$ is finite over X ; there then exists an algebraic space which is a geometric quotient of X by R and a uniform categorical quotient; moreover, if j is separated, this quotient is separated. In particular, if a flat S -group scheme G acts properly on X , with finite stabilizer (i.e. the morphism $G \times_S X \rightarrow X \times_S X$, $(g, x) \mapsto (x, g \cdot x)$ is proper and the stabilizer of the diagonal is finite over X), then there exists a geometric quotient $X \rightarrow X/G$. In the case of a reductive S -group scheme G , this is a result of J. Kollár ([Ko97]).

⁽³⁵⁾**N.D.E.** This hypothesis on X_0 is necessary, cf. N.D.E. (17) in Th. 4.1.

⁽³⁶⁾**N.D.E.** In TDTE III, Th. 6.1, it is stated that $Y \rightarrow S$ is quasi-projective if S is noetherian. The editors have not seen how to deduce this from the local existence of quasi-sections.

⁽³⁷⁾**N.D.E.** The following consequences have been made explicit; see [Ray67a], th. 1 (iii) and the end of the proof of the theorem. Let us also mention that another proof of th. 7.1 in the case of an equivalence relation, based on the existence of Hilbert schemes, is given in [AK80], Th. 2.9; see also [BLR90], §8.2, Th. 12; it is moreover shown there, in this case, that $Y \rightarrow S$ is quasi-projective.

⁽³⁸⁾**N.D.E.** Indeed, if one has constructed such an open neighborhood U_z for every point z closed relative to S , the union of these U_z covers X_0 , since each fiber over S of the closed complement is a noetherian scheme without closed points, hence empty.

Lemma 7.2. *Let T be an affine noetherian scheme, X, Y and Z be T -schemes of finite type, X being quasi-projective over T , and*

$$\begin{array}{ccc} Y & \xrightarrow{u} & X \\ v \downarrow & & \downarrow \\ Z & \dashrightarrow & T \end{array}$$

a diagram of (Sch/T) . Let, on the other hand, z be a point of $v(Y)$ closed relative to T and such that v is flat at the points of $v^{-1}(z)$. Then there exists a closed subscheme F of X such that $u(u^{-1}(F) \cap v^{-1}(z))$ is finite nonempty and the restriction of v to $u^{-1}(F)$ is flat at the points of $v^{-1}(z)$.

Let $T = \text{Spec } A$. One may suppose X of the form $\text{Proj } \mathcal{S}$, where $\mathcal{S}$ is the symmetric algebra of a finitely generated A -module E .

If $u(v^{-1}(z))$ is finite, one can choose $F = X$. Otherwise, denote by $y_1, \dots, y_n$ the points of the fiber $v^{-1}(z)$ associated to its structure sheaf $\mathcal{O}_{v^{-1}(z)}$ (the y_i are thus such that, if $\mathcal{O}_i$ denotes the local ring of $v^{-1}(z)$ at y_i , the maximal ideal of $\mathcal{O}_i$ consists of zero divisors). If t is the image of z in T , $u(v^{-1}(z))$ is an infinite constructible subset of the fiber of t in X . There therefore exists a point x closed in this fiber, belonging to $u(v^{-1}(z))$ and distinct from $u(y_1), \dots, u(y_n)$. Then $X - \{x\}$ is an open neighborhood of $u(y_1), \dots, u(y_n)$, and therefore contains an open neighborhood of the form $D_+(f)$, where f is a homogeneous element of degree d of $\mathcal{S}$ (the notation is that of EGA II, §2.3).

Consequently, the closed subscheme $X_1 = V_+(f)$ defined by f contains x and avoids the points $u(y_1), \dots, u(y_n)$. It follows obviously that the inverse image $Y_1 = u^{-1}(V_+(f))$ of this subscheme is distinct from Y and meets $v^{-1}(z)$. We shall even show that the restriction v_1 of v to Y_1 is flat at the points of $v^{-1}(z)$; if $u(v_1^{-1}(z))$ is finite, one will therefore only have to choose $F = X_1$; otherwise, one repeats the argument just developed, replacing Y by Y_1 , v by v_1 and u by the morphism u_1 induced on Y_1 by u ; in this way one obtains a decreasing sequence $X, X_1, \dots$ of closed subschemes of X ; since such a sequence terminates, $u(u^{-1}(X_n) \cap v^{-1}(z))$ will be finite nonempty for some n , and one chooses $F = X_n$.

It therefore remains to show that v_1 is flat at the points of $v^{-1}(z)$; let, then, y be a point of Y_1 above z , $\mathcal{O}_y$ the local ring of y in Y , $\overline{\mathcal{O}}_y$ the local ring of y in $v^{-1}(z)$, and $\mathcal{O}_{v(y)}$ the local ring of $v(y)$ in Z . If $g \in \mathcal{S}_1$ is such that $D_+(g)$ is a neighborhood of $u(y)$ in X , let φ be the image of f/g^d in $\mathcal{O}_y$, and $\overline{\varphi}$ its image in $\overline{\mathcal{O}}_y$. It follows from the construction of f that $\overline{\varphi}$ is not a zero divisor in $\overline{\mathcal{O}}_y$; since $\mathcal{O}_y$ is flat over $\mathcal{O}_z$, φ is not a zero divisor in $\mathcal{O}_y$ and $\mathcal{O}_y/\mathcal{O}_y\varphi$ is flat over $\mathcal{O}_z$ (SGA 1, IV 5.7). Now $\mathcal{O}_y/\mathcal{O}_y\varphi$ is none other than the local ring of y in Y_1 .

Lemma 7.3. *Retain the notation and hypotheses of 7.1. Every point z of X_0 closed relative to S then has a saturated open neighborhood U_z such that the groupoid induced by X_* on U_z has a quasi-section.*

The statement being local on S , one can suppose S affine noetherian and apply the preceding lemma to the diagram

$$\begin{array}{ccc} X_1 & \xrightarrow{d_0} & X_0 \\ d_1 \downarrow & & \downarrow \\ X_0 & \dashrightarrow & S \end{array}$$

of (Sch/S) . Let, then, F be a closed subscheme of X_0 such that $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite nonempty and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $d_1^{-1}(z)$. Denote by F_1 and F_2 the inverse images of F under d_0 and $d_0 d'_0 = d_0 d'_1$, and by $\tilde{d}_0, \tilde{d}_1$, etc., the morphisms

induced by d_0, d_1 , etc. One therefore has a commutative diagram

$$\begin{array}{ccccc}
 F_2 & \xrightarrow{\tilde{d}'_1} & F_1 & \xrightarrow{\tilde{d}_0} & F \\
 \tilde{d}'_2 \downarrow & & \tilde{d}_0 \downarrow & & \tilde{q} \searrow \\
 X_1 & \xrightarrow[d_0]{d_1} & X_0 & \dashrightarrow & S,
 \end{array}$$

where the two squares on the left are cartesian and the first row exact (cf. (0,1,2), §1), and where q and $\tilde{q}$ denote the structure morphisms.

Let us first show that there are only finitely many points of F_1 above z .⁽³⁹⁾ Indeed, let s be the image of z in S ; since F is of finite type over S , the fiber $\tilde{q}^{-1}(s)$ is a noetherian scheme. On the other hand, since $\tilde{d}_0$ is proper, $\tilde{d}_0(\tilde{d}_1^{-1}(z))$ is a closed subscheme of $\tilde{q}^{-1}(s)$ consisting of finitely many points. Consequently (cf. EGA I, 6.2.2), the points of this set are closed in $\tilde{q}^{-1}(s)$ and also (since F is closed in X_0) in the fiber $q^{-1}(s)$ of s in X_0 . Let y be one of these points; since the fiber $q^{-1}(s)$ is of finite type over $\kappa(s)$, it contains affine open neighborhoods $\text{Spec } B$ and $\text{Spec } C$ of y and z respectively, where B and C are finitely generated $\kappa(s)$ -algebras. Then y and z correspond to maximal ideals $\mathfrak{p} \subset B$ and $\mathfrak{q} \subset C$, the fields $B/\mathfrak{p}$ and $C/\mathfrak{q}$ have finite degree over $\kappa(s)$, and hence $(B/\mathfrak{p}) \otimes_{\kappa(s)} (C/\mathfrak{q})$ is a finite-dimensional $\kappa(s)$ -algebra whose maximal ideals correspond exactly to the points of $X_0 \times_S X_0$ whose second (resp. first) projection is z (resp. y). There are therefore only finitely many points u of $X_0 \times_S X_0$ whose second projection is z and whose first projection belongs to $\tilde{d}_0(\tilde{d}_1^{-1}(z))$. Finally, since $X_1 \rightarrow X_0 \times_S X_0$ has finite fibers, such a point u comes from finitely many points of X_1 , whence the desired assertion.

The morphism $\tilde{d}_1$ is therefore quasi-finite and flat at the points of F_1 above z . Since $\tilde{d}_1$ is of finite type, by SGA 1, IV 6.10 and EGA III, 4.4.10,⁽⁴⁰⁾ the set Φ of points of F_1 where $\tilde{d}_1$ is not both flat and quasi-finite is closed in F_1 , hence in X_1 (since F_1 is closed in X_1). Since d_1 is proper, $\tilde{d}_1(\Phi)$ is closed, and it does not contain z by what precedes. Put $W = \tilde{d}_1(F_1) - \tilde{d}_1(\Phi)$. Then the restriction of $\tilde{d}_1$ to $\tilde{d}_1^{-1}(W)$ is⁽⁴¹⁾ of finite presentation (in view of the noetherian hypotheses), flat, proper and quasi-finite, hence finite, locally free and open, by EGA III, 4.4.2 and EGA IV₂, 2.1.12 and 2.4.6. Consequently, $\tilde{d}_1(F_1)$ is a neighborhood of z , and W is the largest open subset of X_0 contained in $\tilde{d}_1(F_1)$ above which $\tilde{d}_1$ is both flat and quasi-finite.

We shall see in Lemma 7.4 that the inverse images of Φ under $\tilde{d}'_1$ and $\tilde{d}'_0$ are both identified with the set of points of F_2 where $\tilde{d}'_2$ is not both flat and quasi-finite. It follows that $\tilde{d}_0^{-1}(W) = \tilde{d}'_2(F_2) - \tilde{d}'_2(\tilde{d}'_0^{-1}\Phi)$ coincides with $\tilde{d}_1^{-1}(W) = \tilde{d}'_2(F_2) - \tilde{d}'_2(\tilde{d}'_1^{-1}\Phi)$, that is, W is saturated. Consequently, if one puts $W_1 = \tilde{d}_1^{-1}(W)$, the equality $\tilde{d}_0^{-1}(W) = \tilde{d}_1^{-1}(W)$ implies $\tilde{d}'_2^{-1}\tilde{d}_0^{-1}(W) = \tilde{d}'_2^{-1}\tilde{d}_1^{-1}(W)$, that is, $\tilde{d}'_0^{-1}(W_1) = \tilde{d}'_1^{-1}(W_1)$. Since $\tilde{d}_0$ is faithfully flat

⁽³⁹⁾**N.D.E.** Details have been added and the role of the properness hypothesis on d_0 and d_1 in Theorem 7.1 has been brought out. (One may compare with the statement and proof of Theorem 8.1, where this properness hypothesis is omitted.)

⁽⁴⁰⁾**N.D.E.** If $f : X \rightarrow Y$ is a morphism locally of finite type, the set of $x \in X$ isolated in their fiber $f^{-1}(f(x))$ is open in X : in EGA III, 4.4.10, this is deduced, for Y locally noetherian, from Zariski's "Main Theorem"; on the other hand, for arbitrary Y , it follows from Chevalley's semicontinuity theorem (EGA IV₃, 13.1.3 and 13.1.4). Consequently, f is quasi-finite at x if and only if f is of finite type at x and x is isolated in $f^{-1}(f(x))$; this will be used below, cf. N.D.E. (42).

⁽⁴¹⁾**N.D.E.** The original has been expanded in what follows.

and quasi-compact (for d_0 is, like d_1 , surjective, proper and flat), and the square

$$\begin{array}{ccc} F_2 & \xrightarrow{\tilde{d}'_1} & F_1 \\ \tilde{d}'_0 \downarrow & & \downarrow \tilde{d}_0 \\ F_1 & \xrightarrow{\tilde{d}_0} & F \end{array}$$

is cartesian, it follows that W_1 has the form $\tilde{d}_0^{-1}(U)$, where U is an open subset of F (SGA 1, VIII 4.4). This open subset U of F is a quasi-section for the groupoid with base W induced by X_* . One can therefore choose $U_z = W$.

It remains to state Lemma 7.4, whose proof is classical:

Lemma 7.4. *Consider a cartesian square of schemes*

$$\begin{array}{ccc} F_2 & \xrightarrow{v} & F_1 \\ d' \downarrow & & \downarrow d \\ X_1 & \xrightarrow{u} & X_0 \end{array}$$

and let x be a point of F_2 .

- (i) *If u is flat, d' is flat at x if and only if d is flat at $v(x)$.*
- (ii) *If d is locally of finite type, d' is quasi-finite at x if and only if d is quasi-finite at $v(x)$.⁽⁴²⁾*

We have therefore proved that there exists a covering of X_0 by saturated open subsets W such that the groupoid W_* induced by X_* on W has a quasi-section.⁽⁴³⁾

By Lemma 6.1 and the reductions stated after Theorem 7.1, this implies assertions (i) and (iii) of Theorem 7.1, and the facts that p is surjective and open, and that p and $Y \rightarrow S$ are locally of finite presentation. Moreover, since $X_0 \rightarrow S$ is quasi-projective, hence separated and of finite type, p is separated, and the proof of point (ii) of Lemma 6.1 shows that p and $Y \rightarrow S$ are of finite presentation.

To show that p is proper, it therefore remains to show that it is universally closed. Since the assertion is local on Y , one can work on a saturated open subset W such that the groupoid W_* induced by X_* on W has a quasi-section U (since X_0 is covered by such open subsets). Returning to the notation of 6.a), one has a commutative diagram

$$\begin{array}{ccccc} & & W & \xleftarrow{v} & V & \xrightarrow{u} & U \\ & & \searrow & & \downarrow & & \swarrow \\ & & & & Z & & \\ & & p \searrow & & \downarrow r & & \swarrow q \\ & & & & Z, & & \end{array}$$

where Z is an open subset of Y , all arrows are surjective, and q is integral. Moreover, by hypothesis, $d_0 : X_1 \rightarrow X_0$ is proper, so u , obtained from it by base change, is also so. Consequently, r is universally closed, and hence so is p , since v is surjective.

Finally, p being surjective and universally closed, and X_0 quasi-projective, hence separated, the diagonal $\Delta_{Y/S}(Y)$ is closed in $Y \times_S Y$, being the image under $p \times p$ of the diagonal $\Delta_{X_0/S}(X_0)$. Thus Y is separated over S . This completes the proof of 7.1 (ii).

⁽⁴²⁾**N.D.E.** The conditions are sufficient, by base change (cf. EGA II, 6.2.4 (iii) and EGA IV₂, 2.1.4). Conversely, put $y = d'(x)$ and $z = u(y) = d(v(x))$, and suppose d' flat at x and u (hence also v) flat. Then $\mathcal{O}_{v(x)} \rightarrow \mathcal{O}_x$ is faithfully flat, as is $\mathcal{O}_z \rightarrow \mathcal{O}_y \rightarrow \mathcal{O}_x$. Consequently $\mathcal{O}_z \rightarrow \mathcal{O}_{v(x)}$ is faithfully flat (cf. EGA IV₂, 2.2.11 (iv)). Finally, suppose d locally of finite type and d' quasi-finite at x . Then $v(x)$ is isolated in its fiber $d^{-1}(z)$, since x is isolated in its fiber $d'^{-1}(y) = d^{-1}(z) \otimes_{\kappa(z)} \kappa(y)$. Thus, by Chevalley's semicontinuity theorem, there exists an open neighborhood of $v(x)$ every point of which is isolated in its fiber (EGA IV₃, 13.1.3 and 13.1.4), so d is quasi-finite at $v(x)$.

⁽⁴³⁾**N.D.E.** What follows has been modified, taking advantage of the additions made to Lemma 6.1.

The assertions to be proved in 7.1 (iv) are local on Y ; since X_0 is covered by the saturated open subsets U_z , it suffices to verify these assertions by replacing X and Y by U_z and $V = p(U_z)$. As already seen at the beginning of the proof of 7.1, it follows from Lemmas 1.1, 1.2 and 6.1 (i) that $(V_z, p|_{U_z})$ is the cokernel in (Sch) and in (Esp.An) of the groupoid induced by X_* on U_z . Now the hypothesis that $d = (d_0, d_1)$ is a monomorphism is preserved by the base change $U_z \rightarrow X_0$. Consequently, the first two assertions of 7.1 (iv) follow from 6.1 (iv).

Finally, let us show the consequences noted at the end of point (iv) (cf. [Ray67a], th. 1 (iii)). By hypothesis, the groupoid X_* comes from an equivalence relation $R \rightarrow X_0 \times X_0$, and one has established that R is effective (cf. Exp. IV, 3.3.2) and that $p : X_0 \rightarrow Y = X_0/R$ is faithfully flat and of finite presentation. Consequently, denoting by (M) the family of faithfully flat morphisms locally of finite presentation, R is (M) -effective. Thus, by Exp. IV, 6.3.3, (Y, p) represents the quotient sheaf of X_0 by R for the fppf topology, and the assertions concerning base change follow from IV, 3.4.3.1.

8. Passage to the quotient by a flat groupoid that is not necessarily proper

Theorem 8.1. ⁽⁴⁴⁾ *Let S be a noetherian scheme and*

$$X_2 \begin{array}{c} \xrightarrow{d_2'} \\ \xrightarrow{d_1'} \\ \xrightarrow{d_0'} \end{array} X_1 \xrightarrow{d_0} X_0$$

a (Sch/ S)-groupoid such that d_1 is flat and of finite type, X_0 is of finite type over S , and the morphism $X_1 \rightarrow X_0 \times_S X_0$ with components d_0 and d_1 is quasi-finite.

There then exists a dense saturated open subset W of X_0 satisfying the following properties:

- (i) *If $W_2 \xrightarrow{w_2'} W_1 \xrightarrow{w_1'} W$ is the groupoid induced by X_* on W , (w_0, w_1) has a cokernel (V, p) in (Sch/ S); moreover (V, p) is a cokernel of (w_0, w_1) in the category of all ringed spaces.*
- (ii) *p is surjective and open.*
- (ii') *p and $V \rightarrow S$ are of finite presentation.*
- (iii) *The morphism $W_1 \rightarrow W \times_V W$ with components w_0 and w_1 is surjective.*

⁽⁴⁴⁾**N.D.E.** There exists a largest open subset W of X_0 satisfying the conclusions of the theorem. Indeed, let W be an open subset as in the theorem and $W^\#$ a dense saturated open subset contained in W . Since p is open, $V^\# = p(W^\#)$ is an open subset of V , and $W^\# = p^{-1}(V^\#)$, since $W^\#$ is saturated. By Lemma 1.1, $V^\#$ is a cokernel for the groupoid induced on $W^\#$. Thus one can glue along their intersection $W^\#$ two open subsets W and W' satisfying the conclusions of the theorem, and conditions (i), (ii), (iii), (iv), as well as the fact that p and $V \rightarrow S$ are locally of finite presentation, are preserved. Conclusion (ii') follows, as in the proof of 6.1 (ii), from the hypothesis that X_0 is of finite type over noetherian S .

Furthermore, Lemmas 1.1 and 1.2 also show that the union of all saturated open subsets U of X_0 such that the open subset $p(U)$ of Y is a scheme and $p|_U : U \rightarrow p(U)$ is a morphism of schemes is the largest saturated open subset Ω of X_0 satisfying condition (i) of 8.1. Theorem 8.1 shows that Ω contains a dense open subset W , but it is not immediate that Ω satisfies properties (ii) to (iv).

On this subject, the reader may consult [Ray67a], [Ray67b] and appendix I of [An73], which give more precise results and study the question of representability of the quotient fppf S -sheaf $\widetilde{X/R}$ (where R denotes the groupoid X_* with base $X = X_0$), all this under weaker hypotheses (S an arbitrary scheme, X a scheme locally of finite type over S , and R an S -groupoid with base X such that d_0 (and hence d_1) is flat and of finite presentation). Let us mention in particular the following results. If $\widetilde{X/R}$ is represented by an S -scheme Y , then Y is also the cokernel in the category (Esp.An). The converse is false in general (cf. example 0.4 of [Mum65], Chap. 0, §3, cited in [Ray67a], Rem. 1), but is true if $d = (d_0, d_1)$ is an immersion. Under this hypothesis, the morphism $p : \Omega \rightarrow Z := \Omega/R_\Omega$ is faithfully flat and of finite presentation; if moreover S is locally noetherian, a point x of codimension 1 in X belongs to Ω if and only if the graph of the groupoid induced on $\text{Spec}(\mathcal{O}_{X,x})$ is closed. For all this, see [Ray67a], Prop. 1, [Ray67b], Prop. 1 and Theorems 2, 1 and 4, and [An73], Theorems 5 and 6 on pages 66–67 and Prop. 3.3.1 on page 49. (See also, in the case of an action of an algebraic group over an algebraically closed field k , the article [DR81].)

- (iv) If (d_0, d_1) is an equivalence pair, $W_1 \longrightarrow W \times_V W$ is an isomorphism and p is faithfully flat.

We shall show that one can choose W so that the (Sch/S) -groupoid W_* induced by X_* has a quasi-section (cf. §7). Theorem 8.1 will then follow from Lemma 6.1.

Let us provisionally admit that, for every point $z \in X_0$ closed relative to S (cf. §7), there exists a saturated open subset W_z which has a quasi-section and meets all irreducible components of X_0 passing through z . Then the exterior $X_0 - \overline{W}_z$ of W_z in X_0 is saturated (for the saturation $d_1(d_0^{-1}(X_0 - \overline{W}_z))$ of this exterior is open and does not meet W_z). If this exterior is nonempty, one can choose in it a point z' closed relative to S and associate to z' an open subset $W_{z'}$ as above; one can moreover suppose $W_{z'}$ contained in $X_0 - \overline{W}_z$; then W_z and $W_{z'}$ are disjoint and the groupoid induced by X_* on $W_z \cup W_{z'}$ has a quasi-section. The process must terminate because X_0 has only finitely many irreducible components. It therefore remains to construct W_z .

For this one may suppose S affine; in this case, let y be a point of X_1 such that $d_1(y) = z$, X an affine open subset of X_0 containing $d_0(y)$, Y the inverse image of X in X_1 under d_0 , and finally $u : Y \longrightarrow X$ and $v : Y \longrightarrow X_0$ the morphisms induced by d_0 and d_1 . Since X is affine, hence quasi-projective, one can apply Lemma 7.2: there is therefore a subscheme F of X_0 such that $d_0^{-1}(F) \cap d_1^{-1}(z)$ is nonempty, $d_0(d_0^{-1}(F) \cap d_1^{-1}(z))$ is finite, and the restriction of d_1 to $d_0^{-1}(F)$ is flat at the points of $v^{-1}(z)$. This allows us to return to the notation of Lemma 7.3, denoting by F_1 and F_2 the inverse images of F in X_1 and X_2 , etc.

$$\begin{array}{ccccc} F_2 & \xrightarrow{\tilde{d}'_1} & F_1 & \xrightarrow{\tilde{d}_0} & F \\ \tilde{d}'_2 \downarrow & & \tilde{d}'_0 \downarrow & & \tilde{d}_1 \downarrow \\ X_1 & \xrightarrow[d_0]{d_1} & X_0 & & \end{array} .$$

One then shows, as in 7.3, that $\tilde{d}_1$ is quasi-finite at the points of $\tilde{d}_1^{-1}(z)$, so it is natural to consider the open subset F'_1 of F_1 consisting of the points where $\tilde{d}_1$ is both flat and quasi-finite. By 7.4, the two inverse images of F'_1 under $\tilde{d}'_1$ and $\tilde{d}'_0$ consist of the points of F_2 where $\tilde{d}'_2$ is flat and quasi-finite, so these two inverse images coincide and F'_1 has the form $\tilde{d}_0^{-1}(F')$, where F' is an open subset of F (SGA 1, VIII 4.4). By replacing F by F' if necessary, one can therefore suppose $\tilde{d}_1$ quasi-finite and flat. In this case, denote by W_z the largest open subset of $\tilde{d}_1(F_1)$ above which $\tilde{d}_1$ is finite and flat.

This open subset W_z does not necessarily contain z , but it contains the generic points of the irreducible components of X_0 passing through z .⁽⁴⁵⁾ Since d_0 (resp. d_1) is faithfully flat and of finite presentation (hence open), it follows from SGA 1, VIII 5.7 that $d_0^{-1}(W_z)$ and $d_1^{-1}(W_z)$ both coincide with the largest open subset of $\tilde{d}'_2(F_2)$ above which $\tilde{d}'_2$ is finite and flat. Consequently, one sees, as in 7.3, that the two inverse images of $\tilde{d}_1^{-1}(W_z)$ under $\tilde{d}'_0$ and $\tilde{d}'_1$ coincide, hence that $\tilde{d}_1^{-1}(W_z)$ has the form $\tilde{d}_0^{-1}(U)$, where U is an open subset of F which is a quasi-section for the groupoid induced by X_* on W_z .

9. Elimination of the noetherian hypotheses in Theorem 7.1

a) Return to the notation and hypotheses of Lemma 6.1, and let $\pi : S' \longrightarrow S$ be an arbitrary base change. Denote by $f' : X' \longrightarrow Y'$ the morphism of S' -schemes obtained by the extension π of the base from a morphism of S -schemes $f : X \longrightarrow Y$. With this convention, $p' : X'_0 \longrightarrow Y'$ is surjective, as is the morphism $X'_1 \longrightarrow X'_0 \times_{Y'} X'_0$ with components d'_0 and d'_1 . The underlying set of Y' is therefore identified with the quotient of the underlying set of

⁽⁴⁵⁾**N.D.E.** Indeed, let η be such a generic point. The hypotheses imply that $\mathcal{O}_{X_0, \eta}$ is an artinian local ring, and $\mathcal{O}_{F_1, \eta}$ a finitely generated $\mathcal{O}_{X_0, \eta}$ -module. Thus, by SGA 1, VIII 6.5, there exists an open neighborhood of η above which d_1 is finite.

X'_0 by the equivalence relation defined on X'_0 by the S' -groupoid X'_* . Moreover, $q' : U' \rightarrow Y'$ is integral surjective, so the topology of Y' is the quotient topology of that of U' , and thus also of that of X'_0 (cf. the proof in §6.c).

On the other hand, it is clear that U' is a quasi-section of the S' -groupoid X'_* , to which one can therefore apply Lemma 6.1. In particular, X'_* has a cokernel (Y_1, p_1) , and the underlying topological space of Y_1 is obtained from the underlying topological space of X'_0 by identifying the points equivalent for the relation defined by X'_* . It follows that the canonical morphism $Y_1 \rightarrow Y'$ is a homeomorphism; I even say that $Y_1 \rightarrow Y'$ is a universal homeomorphism: indeed, if S'' is over S' , let Y_2 be the cokernel of $(d_0 \times_S S'', d_1 \times_S S'')$. By what precedes, applied to the base changes $S'' \rightarrow S'$ and $S'' \rightarrow S$,

$$Y_2 \rightarrow Y_1 \times_{S'} S'' \quad \text{and} \quad Y_2 \rightarrow Y \times_S S'' \simeq Y' \times_{S'} S''$$

are homeomorphisms, so the same holds for $Y_1 \times_{S'} S'' \rightarrow Y' \times_{S'} S''$.

b) Analogous remarks obviously apply to the case where the groupoid X_* “locally” has quasi-sections (cf. the proof of Theorem 7.1).

⁽⁴⁶⁾For example, one has the following theorem:

Theorem 9.0. *Let S be an arbitrary scheme and $X_2 \xrightarrow{d'_j} X_1 \xrightarrow{d_i} X_0$ a (Sch/S) -groupoid such that: X_0 is of finite presentation and quasi-projective over S , d_1 is of finite presentation, proper and flat, and the morphism $d_0 \boxtimes d_1 : X_1 \rightarrow X_0 \times_S X_0$ is quasi-finite. Then:*

- (1) *Every point x of X_0 has an open neighborhood W which is saturated and such that the groupoid induced by X_* on W has a quasi-section.*
- (2) *Let (Y, p) be the cokernel of (d_0, d_1) in the category of all ringed spaces. Then Y is a scheme, p a morphism of schemes, and (Y, p) is a cokernel of (d_0, d_1) in (Sch/S) .*
- (3) *p is surjective, open and universally closed.*
- (4) *The morphism $d : X_1 \rightarrow X_0 \times_Y X_0$ with components d_0 and d_1 is surjective.*
- (5) *If (d_0, d_1) is an equivalence pair, then:*
 - (a) *$d : X_1 \rightarrow X_0 \times_Y X_0$ is an isomorphism and p is faithfully flat.*
 - (b) *p and $Y \rightarrow S$ are of finite presentation, and (Y, p) is a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology.*

PROOF. For (1), the question is local on S , so one can suppose $S = \text{Spec } B$ affine. There then exist a ring A finitely generated over $\mathbf{Z}$, a morphism $S \rightarrow T = \text{Spec } A$ and a (Sch/T) -groupoid Z_* such that X_* is identified with $Z_* \times_T S$ (cf. EGA IV₃, 8.8.3, applied to $S_0 = \text{Spec } \mathbf{Z}$ and $S_i = \text{Spec } A_i$, the A_i ranging over the finitely generated $\mathbf{Z}$ -subalgebras of B). Moreover, one can suppose that Z_* satisfies the conditions of Theorem 7.1 (cf. EGA IV₃, 8.10.5). Consequently, Z_* “locally” has quasi-sections.

The same therefore holds for X_* by a), and assertions (2), (3), (4) and (5) (a) follow from 6.1, as in the proof of 7.1.

c) Let us show that $Y \rightarrow S$ is of finite presentation.⁽⁴⁷⁾ By hypothesis, $(d_0^{X_*}, d_1^{X_*})$ is an equivalence pair, that is, $d^{X_*} : X_1 \rightarrow X_0 \times_S X_0$ is a monomorphism. By EGA IV₃, 8.10.5, one can suppose, enlarging A if necessary, that $d^{Z_*} : Z_1 \rightarrow Z_0 \times_T Z_0$ is a monomorphism. Since $T = \text{Spec } A$, with A noetherian, it then follows from Theorem 7.1 that the groupoid Z_* has a cokernel (Q, q) in (Sch/T) , that q and $Q \rightarrow T$ are of finite presentation, and moreover that $q : Z_0 \rightarrow Q$ is faithfully flat and d^{Z_*} induces an isomorphism $Z_1 \xrightarrow{\sim} Z_0 \times_Q Z_0$. Put $Q_S = Q \times_T S$.

Since $X_i \simeq Z_i \times_T S$, one therefore obtains an isomorphism:

$$d^{Z_*} \times_T S : X_1 \xrightarrow{\sim} X_0 \times_{Q_S} X_0.$$

⁽⁴⁶⁾N.D.E. What follows has been expanded to bring out Theorem 9.0 below.

⁽⁴⁷⁾N.D.E. The original states that this follows from Proposition 9.1 below. We have been unable to reconstruct this argument. The proof that follows was indicated to us by O. Gabber.

Denote its inverse by φ , and let π be the canonical morphism

$$X_0 \times_Y X_0 \longrightarrow X_0 \times_{Q_S} X_0.$$

Then $\varphi \circ \pi$ is the inverse of $d_0 \boxtimes d_1 : X_1 \xrightarrow{\sim} X_0 \times_Y X_0$. It follows that the equivalence relation defined by X_* , that is, the monomorphism

$$X_1 \xrightarrow{\sim} X_0 \times_Y X_0 \hookrightarrow X_0 \times_S X_0,$$

is identified with the equivalence relation $R(q_S)$ defined by the morphism $q_S : X_0 \longrightarrow Q_S$. Since the latter is faithfully flat and of finite presentation, hence an effective universal epimorphism, $R(q_S)$ has quotient Q_S (cf. IV 3.3.2). Thus $Y \simeq Q \times_T S$, so $Y \longrightarrow S$ and p are of finite presentation. Moreover, by IV 6.3.3, (Y, p) is also a cokernel of (d_0, d_1) in the category of sheaves for the fppf topology. $\square$

Proposition 9.1. *Consider morphisms of schemes*

$$X_0 \xrightarrow{p} Y \xrightarrow{q} S$$

such that qp is of finite type (resp. of finite presentation) and p faithfully flat of finite presentation. Then q is of finite type (resp. of finite presentation).⁽⁴⁸⁾

Since p is surjective and qp quasi-compact, q is quasi-compact. One can therefore suppose S, Y and X_0 affine with rings A, B, C . One has $B = \varinjlim B_i$, where the B_i range over the finitely generated A -subalgebras of B . Since C is of finite presentation over B , there exist an index i_0 , a finitely presented B_{i_0} -algebra C_{i_0} and an isomorphism $C \simeq C_{i_0} \otimes_{B_{i_0}} B$; if one puts $C_i = C_{i_0} \otimes_{B_{i_0}} B_i$ for $i \geq i_0$, one therefore has $C \simeq C_i \otimes_{B_i} B$.

$$\begin{array}{ccc} B & \longrightarrow & C \\ \uparrow & & \uparrow \\ B_i & \longrightarrow & C_i \\ \uparrow & & \\ A & & \end{array}$$

Since C is faithfully flat over B , one deduces from EGA IV₃, 11.2.6 and 8.10.5 (vi) the existence of an $i_1 \geq i_0$ such that C_{i_1} is faithfully flat over B_{i_1} ; consequently C_i is faithfully flat over B_i for $i \geq i_1$. For $i \geq i_1$, the canonical map $C_i \longrightarrow C$ is then injective, for it is obtained from $B_i \longrightarrow B$ by faithfully flat extension of the base.

If C is of finite type over A , it follows that there exists an index $j \geq i_1$ such that $C_j = C$, whence $B_j = B$, since C_j is faithfully flat over B_j . Consequently B is of finite type over A .

Now suppose C of finite presentation over A . By what precedes, B is of finite type over A , hence of the form $\overline{B}/I$, where $\overline{B}$ is a polynomial algebra over A in finitely many indeterminates and I an ideal of $\overline{B}$. Then I is the union of its finitely generated subideals I_α ; whence $B = \varinjlim B_\alpha$ with $B_\alpha = \overline{B}/I_\alpha$. Proceeding as above, there exist an index α_0 , a finitely presented B_{α_0} -algebra C_{α_0} and an isomorphism $C \simeq C_{\alpha_0} \otimes_{B_{\alpha_0}} B$. For $\alpha \geq \alpha_0$, one again puts $C_\alpha = C_{\alpha_0} \otimes_{B_{\alpha_0}} B_\alpha$, so that $C \simeq C_\alpha \otimes_{B_\alpha} B$ for $\alpha \geq \alpha_0$. Again by EGA IV₃, 11.2.6 and 8.10.5 (vi), one concludes as above that C_α is faithfully flat over B_α for sufficiently large α . In this case, the kernel of the map $C_\alpha \longrightarrow C$ (resp. $C_\alpha \longrightarrow C_\beta$ for $\beta \geq \alpha$) is identified with $C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ (resp. $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha)$).

Since C_α and C are of finite presentation over A and $C_\alpha \longrightarrow C$ is surjective, $C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ is a finitely generated ideal⁽⁴⁹⁾ and is the union of the ideals $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha)$. One therefore has $C_\alpha \otimes_{B_\alpha} (I_\beta/I_\alpha) = C_\alpha \otimes_{B_\alpha} (I/I_\alpha)$ for sufficiently large β , whence also $I_\beta = I$ (for C_α is faithfully flat over B_α); thus B is of finite presentation over A .

⁽⁴⁸⁾Cf. EGA IV₄, 17.7.5 for a more general result.

⁽⁴⁹⁾**N.D.E.** Cf. EGA IV₁, 1.4.4.

10. Supplement: quotients by a group scheme

Paragraphs 10.2–10.4 below, written following indications of M. Raynaud, aim to apply the preceding theorems to the case of an action of a group scheme. For the reader's convenience, we have begun by reproducing, in §10.1, statements 2.1 to 2.3 of Exp. XVI.

10.1. — Representability theorems for quotients. — Let us first “recall” the following result:

Theorem 10.1.1. *Let S be a scheme, X and Y two S -schemes, and $f : X \rightarrow Y$ an S -morphism. Suppose one is in one of the following two cases:*

- α) The morphism f is locally of finite presentation.*
- β) The scheme S is locally noetherian and X is locally of finite type over S .*

Then the following conditions are equivalent:

- (i) *There exist an S -scheme X' and a factorization of f :*

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S -morphism locally of finite presentation and f'' is a monomorphism.

- (ii) *The (first) projection*

$$p_1 : X \times_Y X \rightarrow X$$

is a flat morphism.

Moreover, if the preceding conditions hold, (X', f') is a quotient of X by the equivalence relation defined by f (for the fppf topology), so the factorization $f = f'' \circ f'$ of i) is unique up to isomorphism.

The case Y locally noetherian, X of finite type over Y , is treated in [Mur65], cor. 2 of th. 2. We shall see that one can reduce to this case.

Let us first make some remarks:

- a) The implication (i) $\Rightarrow$ (ii) is trivial. Indeed, the first projection

$$p'_1 : X \times_{X'} X \rightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism, since f'' is a monomorphism, and p'_1 is flat, since f' is flat; hence p_1 is flat.

b) The assertions of 10.1.1 are local on Y (hence local on S); they are also local on X , as follows easily from the fact that a flat morphism locally of finite presentation is open (EGA IV₃, 11.3.1).

c) Under the hypotheses of 10.1.1 α), in view of what precedes, we are reduced to the case where X and Y are affine and f of finite presentation. By replacing S by Y if necessary, one can suppose X and Y of finite presentation over S . One then reduces to the case S noetherian by EGA IV₃, 11.2.6.

d) Under the hypotheses of 10.1.1 β), one can suppose S, X, Y affine, S noetherian and X of finite type over S . Regard Y as a filtered projective limit of affine schemes Y_i of finite type over S . The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has $X \times_{Y_i} X = X \times_Y X$ for sufficiently large i , so that $f_i : X \xrightarrow{f} Y \rightarrow Y_i$ satisfies the hypotheses of 10.1.1 ii) if f does. Since the equivalence relation defined by f on X coincides with that defined by f_i , it is clear that it suffices to prove ii) $\Rightarrow$ i) for f_i , which reduces us to the case where Y is of finite type over S .

Application to group schemes. Let S be a scheme and G an S -group scheme locally of finite presentation over S , acting (on the left) on an S -scheme X . If $X \rightarrow S$ has a section ξ , recall that the stabilizer $\text{Stab}_G(\xi)$ is representable by a subgroup scheme of G (cf. I, 2.3.3).

Theorem 10.1.2. *Let S be a scheme and G an S -group scheme locally of finite presentation over S , acting on an S -scheme X .*

Suppose $X \rightarrow S$ has a section ξ whose stabilizer H in G is flat over S . If one of the hypotheses below holds:

- a) X is locally of finite type over S ,
- b) S is locally noetherian,

then the quotient fppf sheaf G/H is representable by an S -scheme locally of finite presentation over S , and the S -morphism

$$f : G \rightarrow X, \quad g \mapsto g \cdot \xi$$

factors as:

$$\begin{array}{ccc} G & & \\ p \downarrow & \searrow f & \\ G/H & \xrightarrow{i} & X, \end{array}$$

where p is the canonical projection, a faithfully flat morphism locally of finite presentation, and i is a monomorphism.

PROOF. The morphism f makes G an X -scheme. By definition of the stabilizer of ξ , the morphism

$$G \times_S H \rightarrow G \times_X G, \quad (g, h) \mapsto (g, gh)$$

is an isomorphism. Since H is flat over S , $G \times_S H$ is flat over G , hence the first projection $p_1 : G \times_S G \rightarrow G$ is a flat morphism. Furthermore, if X is locally of finite type over S , f is locally of finite presentation (EGA IV₁, 1.4.3 (v)); otherwise S is supposed locally noetherian. One then need only apply 10.1.1 to the morphism f . It remains to see that G/H is locally of finite presentation over S , but this follows immediately from 9.1. $\square$

Corollary 10.1.3. *Let S be a scheme and $u : G \rightarrow H$ a morphism of S -group schemes. Suppose G locally of finite presentation over S , and either H locally of finite type over S or S locally noetherian.*

Then, if $K = \text{Ker}(u)$ is flat over S , the quotient group G/K is representable by an S -group scheme locally of finite presentation over S , and u factors as:

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \searrow & & \nearrow i \\ & G/K & \end{array}$$

where p is the canonical projection and i a monomorphism.

Proof: one applies 10.1.2 with $X = H$ and ξ the unit section of H .

10.2. — Stabilizer of the diagonal. — Let S be a noetherian scheme, X an S -scheme of finite type, and G a flat S -group scheme of finite type acting on X on the left, i.e. one has an S -action $d_0 : G \times_S X \rightarrow X$. Denote by $d_1 : G \times_S X \rightarrow X$ the projection onto the second factor. Following §2.a), one has the groupoid

$$G \times_S G \times_S X \begin{array}{c} \xrightarrow{\text{pr}_{2,3}} \\ \xrightarrow[-\mu \times X]{G \times d_0} \end{array} G \times_S X \begin{array}{c} \xrightarrow{d_1} \\ \xrightarrow{d_0} \end{array} X,$$

whose cokernel, if it exists, is denoted $G \backslash X$.

Definition 10.2.1. Denote by $F \subset G \times_S X$ the stabilizer of the diagonal section, i.e. the X -scheme defined by the cartesian product

$$\begin{array}{ccc} F & \longrightarrow & X \\ \downarrow & & \downarrow \Delta \\ G \times_S X & \xrightarrow{(d_0, d_1)} & X \times_S X. \end{array}$$

Then F is a subgroup X -scheme of $G \times_S X$. Since $G \times_S X$ is of finite type over noetherian S , hence noetherian, F is of finite type over S and over X (EGA I, 6.3.5 and 6.3.6). Moreover, if $X \rightarrow S$ is separated, F is a closed subgroup X -scheme of $G \times_S X$.

Recall that one says G acts freely on X if the morphism

$$G \times_S X \xrightarrow{(d_0, d_1)} X \times_S X$$

is a monomorphism (cf. Exp. III, 3.2.1). This amounts to saying that F is the trivial group scheme with base X .

10.3. — Case where F is quasi-finite over X . — Since F is of finite type over X , it is quasi-finite over X if and only if the stabilizers of the geometric points of X are finite.

Theorem 10.3.1. ⁽⁵⁰⁾ *Under the hypotheses of 10.2, suppose F quasi-finite over X . Then there exists a dense G -saturated open subset U of X satisfying the following properties:*

- (i) *In (Sch/S) , the cokernel $V = G \backslash U$ exists; moreover, the scheme V is a quotient in the category of ringed spaces.*
- (ii) *$p : U \rightarrow V$ is surjective, open and of finite presentation.*
- (iii) *V is of finite presentation over S .*
- (iv) *The morphism $G \times_S U \rightarrow U \times_V U$, $(g, x) \mapsto (gx, x)$, is surjective.*
- (v) *Further suppose G acts freely on X . Then $U \rightarrow V$ is a (left) G -torsor locally trivial for the fppf topology. In particular, $U \rightarrow V$ is faithfully flat.⁽⁵¹⁾*

PROOF. One has supposed that the morphism $G \times_S X \rightarrow X \times_S X$, $(g, x) \mapsto (gx, x)$, is quasi-finite. Theorem 8.1 therefore applies to the groupoid defined by (X, G) . Thus there exists a dense saturated open subset $U \subset X$ such that the quotient $G \backslash U$ exists; it satisfies properties (i), (ii), (iii).

To establish (iv), recall that G acts freely on X if and only if (d_0, d_1) is an equivalence pair (III 3.2.1). In this case, Theorem 8.1 (iv) shows that the morphism $G \times_S U \rightarrow U \times_V U$ is an isomorphism and p is faithfully flat and of finite presentation. Thus U is a G -torsor with base V , locally trivial for the fppf topology. $\square$

10.4. — Case where F is flat over X . — Denote by

$$d = (d_0, d_1) : G \times_S X \rightarrow X \times_S X$$

the morphism $d(g, x) = (gx, x)$. Recall that the sheaf graph $\tilde{\Gamma}$ of the equivalence relation associated to (X, G) is the fppf S -subsheaf of $X \times_S X$ that is the image of (d_0, d_1) . It is the fppf sheaf associated to the graph functor:

$$T \mapsto \Gamma(T) = \{(x_0, x_1) \in X(T) \times X(T) \mid x_0 \in G(T)x_1\}.$$

Put $G_X = G \times_S X$. For every S -scheme T , one has a surjective map

$$G_X(T) \rightarrow \Gamma(T), \quad (g, x) \mapsto (gx, x),$$

⁽⁵⁰⁾**N.D.E.** Here too there exists a largest open subset U of X satisfying the conclusions of the theorem, cf. N.D.E. (44).

⁽⁵¹⁾**N.D.E.** If one further supposes that G is a reductive S -group scheme and the (free) action of G on X is linearizable, one then knows that $G \backslash X$ is representable and $X \rightarrow G \backslash X$ is a (left) G -torsor. This follows from results of Raynaud and Seshadri and appears in the article [CTS79] (proposition 6.11).

which induces a bijective map

$$\varphi(T) : G_X(T)/F(T) \xrightarrow{\sim} \Gamma(T);$$

indeed, if $(g, x), (g', x') \in G_X(T)$ satisfy $(gx, x) = (g'x', x')$, then $x' = x$ and $g^{-1}g'x = x$, hence $(g^{-1}g'x, x) \in F(T)$, and (g, x) and (g', x) have the same image in $G_X(T)/F_X(T)$.

By definition (cf. IV, 4.4.1 (ii) or the proof of 5.2.1), the quotient sheaf G_X/F is the fppf sheaf associated to the presheaf

$$T \mapsto G_X(T)/F(T) \simeq \Gamma(T).$$

One therefore has an isomorphism of sheaves $\varphi : G_X/F \rightarrow \tilde{\Gamma}$.

Theorem 10.4.1. ⁽⁵²⁾ *Under the hypotheses of 10.2, one has:*

- a) $\tilde{\Gamma}$ is representable if and only if F is flat over X .
- b) Suppose F flat over X . Then the morphisms induced by d_1 and d_0 :

$$G_X/F \begin{matrix} \xrightarrow{\bar{d}_1} \\ \xrightarrow{\bar{d}_0} \end{matrix} X$$

are faithfully flat and of finite presentation.

Proof of a): Suppose the fppf sheaf G_X/F represented by an X -scheme Y . Then, by IV 6.3.3, $p : G_X \rightarrow Y$ is faithfully flat and locally of finite presentation, and the second square of the diagram below is cartesian:

$$\begin{array}{ccccc} F & \longrightarrow & F \times_X G_X & \longrightarrow & G_X \\ \downarrow & & \downarrow & & \downarrow p \\ X & \xrightarrow{e_X} & G_X & \longrightarrow & Y, \end{array}$$

the first square being obtained by base change by the unit section $e_X : X \rightarrow G_X$. Since p is faithfully flat and locally of finite presentation, the same holds for $F \rightarrow X$.

Conversely, suppose F flat over X . Put $X^2 = X \times_S X$. The morphism $d : G_X \rightarrow X^2$ allows one to form the fibered product:

$$\begin{array}{ccc} G_X \times_{X^2} G_X & \longrightarrow & G_X \\ \downarrow & & \downarrow \\ G_X & \longrightarrow & X^2. \end{array}$$

Then the morphism $G_X \times_{X^2} G_X \rightarrow X^2$ is an $F \times_X X^2$ -torsor over X^2 , and is therefore flat and of finite type (since F is). By Theorem 10.1.1, the morphism d factors uniquely:

$$G_X \xrightarrow{\psi} Y \xrightarrow{\tau} X \times_S X,$$

where ψ is faithfully flat (of finite type) and τ is a monomorphism of schemes.

Thus the morphism of sheaves $\psi : G_X \rightarrow Y$ is F -invariant, and one obtains a morphism of sheaves $\bar{\psi} : G_X/F \rightarrow Y$. Furthermore, since ψ is faithfully flat (of finite type), the monomorphism of sheaves τ factors through the image sheaf of d , that is, $\tilde{\Gamma}$. The isomorphism of sheaves $G_X/F \simeq \tilde{\Gamma}$ therefore factors through the monomorphism $Y \rightarrow \tilde{\Gamma}$. One concludes that Y represents G_X/F .

Proof of b): Suppose F flat over X . Then, by a) and its proof, G_X/F is representable, and the morphism $p : G_X \rightarrow G_X/F$ is faithfully flat and of finite presentation. On the other hand, the morphisms $d_i : G_X \rightarrow X$ ($i = 0, 1$) are faithfully flat and of finite presentation by hypothesis. Since $d_i = \bar{d}_i \circ p$, it follows from EGA IV₂, 2.2.13 (iii) and EGA IV₃, 11.3.16 that $\bar{d}_i$ is faithfully flat and of finite presentation.

⁽⁵²⁾**N.D.E.** This is point (2) of Theorem 3 of [Ray67b]. Another proof of th. 10.1.1 is sketched in this Note.

Theorem 10.4.2. ⁽⁵³⁾ *Under the hypotheses of 10.2, suppose F flat over X . Then there exists a dense saturated open subset U of X such that the quotient fppf sheaf $V = G \backslash U$ is an S -scheme of finite type and $U \rightarrow V$ is faithfully flat and of finite presentation.*

PROOF. Theorem 10.4.1 shows that $G_X/F \simeq \tilde{\Gamma}$ is representable. Then the fppf sheaf $G \backslash X$ is identified with the quotient sheaf of

$$G_X/F \xrightarrow[\bar{d}_0]{\bar{d}_1} X.$$

By what precedes, $\bar{d}_i : G_X/F \rightarrow X$ is faithfully flat and of finite presentation ($i = 0, 1$), and the morphism

$$G_X/F \xrightarrow{\sim} \tilde{\Gamma} \hookrightarrow X \times_S X$$

is a monomorphism, that is, $(\bar{d}_0, \bar{d}_1)$ is an equivalence pair. Consequently, Theorem 8.1 applies. There therefore exists a dense saturated open subset U of X such that the quotient fppf sheaf $V = G \backslash U$ is an S -scheme of finite type and $U \rightarrow V$ is faithfully flat and of finite presentation. $\square$

Taking account of the generic flatness theorem (EGA IV₂, 6.9.3), one obtains the

Corollary 10.4.3. *Under the hypotheses of 10.2, suppose X reduced. Then there exists a dense saturated open subset U of X such that the quotient fppf sheaf $G \backslash U$ is an S -scheme of finite type and $U \rightarrow G \backslash U$ is faithfully flat and of finite presentation.*

⁽⁵³⁾**N.D.E.** Here too there exists a largest open subset U of X satisfying the conclusions of the theorem; moreover, a point $x \in X$ of codimension 1 in X belongs to U if and only if the morphism $(G_X/F) \times_X \text{Spec}(\mathcal{O}_{X,x}) \rightarrow \text{Spec}(\mathcal{O}_{X,x}) \times_S \text{Spec}(\mathcal{O}_{X,x})$ is a closed immersion, cf. N.D.E. (44).

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